

Chapter 13

Superconductivity

Abstract This chapter deals with superconductivity. Several experimental observations like the Meissner effect, the critical field, the existence of type I and type II superconductors, the isotope effect and the determination of thermal conductivity and specific heat in the normal and superconducting states are discussed. The thermodynamics of superconductors is discussed in detail. The values of quantities like the free energy, the entropy and the specific heat in the normal and superconducting states are worked out. The concept of the energy gap in the electron energy spectrum is explained. Some of the theories of superconductivity like Londons' theory, the Bardeen–Cooper–Schreiffer theory and the Ginzburg–Landau theory are discussed. The concepts of penetration depth, flux quantization, coherence length and Cooper pairs are explained. The connection between thermal vibrations and superconductivity is emphasized. Tunneling through barriers is an important aspect in superconductivity. Both tunneling of electrons (normal tunneling) and Cooper pairs (Josephson tunneling) are discussed. The AC and DC Josephson effects are discussed. Quantum interference effect and its application (SQUID) are described. In the 1980s, superconductivity was observed in some compounds with transition temperatures above the liquid nitrogen temperature. The structure and properties of some HT superconducting compounds are discussed.

13.1 Introduction

We have noted in earlier chapters that electrical conductivity of solids has a wide range of values; 10^{-22} – 10^{-14} for insulators, 10^{-9} – 10^2 for semiconductors and 10^4 – 10^8 for metals (all in units of $\text{ohm}^{-1} \text{ cm}^{-1}$). In fact the range is still wider, going up to almost infinity in the case of superconductors. But this occurs only at extremely low temperatures. The discovery of superconductors became possible only with the availability of liquid helium. Superconductivity has led to the discovery of several new phenomena, elaborate theories and a number of applications.

In this chapter, we shall start with the discovery of superconductivity and then describe some properties of superconductors. This is followed by a discussion of the main theories. Some applications of superconductivity are discussed. For completeness, we have included a section on high temperature superconductors.

There is a vast literature on superconductivity. Apart from discussions in text books by Kubo and Nagamiya [1], Blakemore [2] and Ashcroft and Mermin [3], Kittel [4], there are books dedicated to superconductivity [5–7].

13.2 Discovery of Superconductivity

Kammerlingh Onnes succeeded in the liquefaction of helium in 1908. This opened up a new low temperature range for studying properties of materials. Onnes took up a study of the electrical resistance of metals at liquid helium temperatures.

Onnes [8] discovered superconductivity while studying the electrical resistance of mercury. The equipment used by Onnes is shown in Fig. 13.1. Purified mercury is introduced in vacuo in a series of U-tubes (A–B) of diameter 0.05 mm. This system of U-tubes is placed in a specially designed cryostat containing liquid

Fig. 13.1 Apparatus used by Kammerlingh Onnes in the discovery of superconductivity

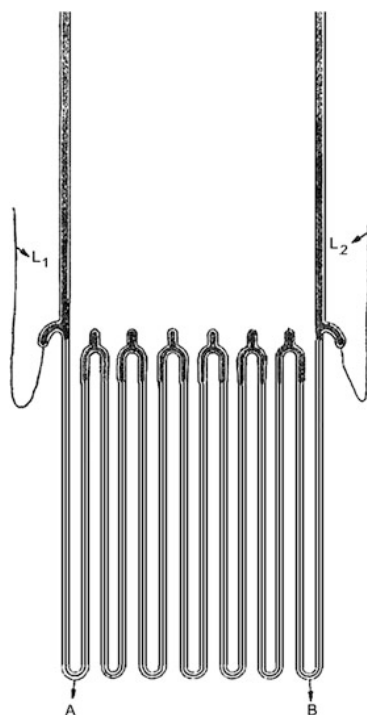
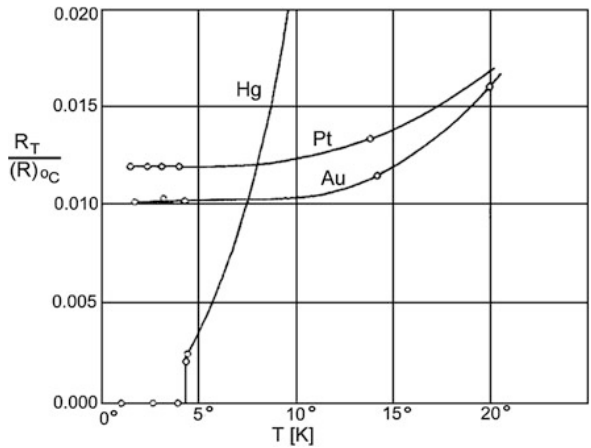


Fig. 13.2 Temperature variation of fractional resistance $[R_T/(R)_{0C}]$ of mercury, platinum and gold at very low temperatures

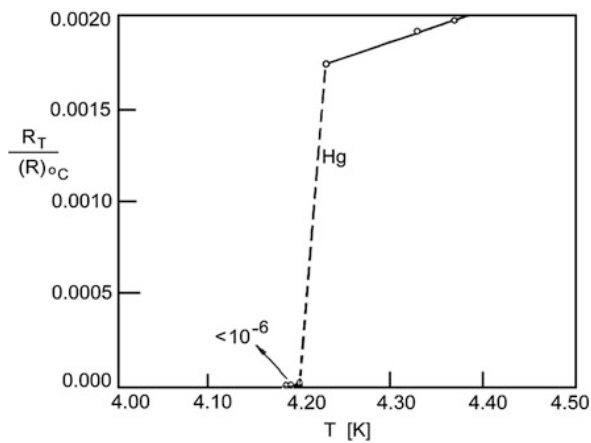


helium where the mercury freezes and becomes a long wire. The ends of the mercury wire are connected to platinum wires L_1 and L_2 to facilitate passing of current and the measurement of potential difference.

The results obtained by Onnes on Hg and a few other metals are shown in Fig. 13.2. The figure shows the electrical resistance (expressed as a fraction of its value at 0 °C) at different temperatures. It can be seen that the resistance of Pt and Au decreases smoothly as the temperature is lowered from about 20 K down to ~ 1.5 K. The resistance of mercury also decreases smoothly (though more steeply than in Pt and Au) till about 5 K but at a temperature of 4.2 K, it abruptly drops almost to zero. In the words of Onnes [9], the resistance which was 1/500 at 4.2 K dropped to ‘a thousand millionth part’ of that at normal temperature.

That the fall in resistance was abrupt and sharp is seen more clearly in Fig. 13.3. In terms of conductivity, the conductivity has reached a value which is a thousand million times that at normal temperature. Onnes concluded that mercury has entered a new state—the state of superconductivity.

Fig. 13.3 Abrupt fall in the fractional resistance $[R_T/(R)_{0C}]$ of mercury



The temperature at which superconductivity appears was called the ‘transition point’ by Onnes. It is generally referred to as the transition temperature T_c . Below T_c , a metal is in the superconducting state and above T_c it behaves like a normal metal.

13.3 Occurrence

Mercury was the first metal in which superconductivity was observed by Kammerlingh Onnes. In the same series of experiments he observed superconductivity in Sn ($T_c = 3.72$ K) and Pb ($T_c = 7.19$ K). Onnes’ discovery of superconductivity led to a study of superconductivity by many other research groups and laboratories and since then superconductivity has been observed in several other metals. A list (not exhaustive) of superconducting metals along with their transition temperatures is given in Table 13.1. Among elements which are superconductors at ambient pressures, niobium has the highest transition temperature ($T_c = 9.25$ K).

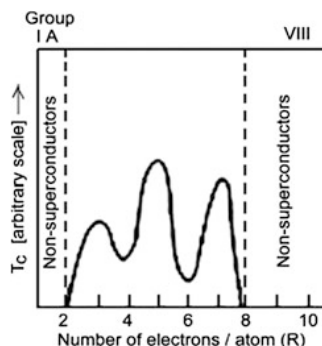
Mathias [10] proposed the following empirical rules for the appearance of superconductivity:

- (i) Superconductivity occurs in metallic elements in which the number of valence electrons per atom (R) has values between 2 and 8 (Fig. 13.4).
- (ii) Elements with odd R have high T_c values and those with even R values have low T_c values.
- (iii) Some elements which are not superconductors in the pure form become superconductors in the alloyed form.
- (iv) Maximum T_c has been observed for alloys and compounds having R close to 5 and 7.

Table 13.1 Values of T_c for some elements which are superconductors at atmospheric pressure

Element	T_c (K)
Niobium	9.25
Lead	7.19
Tantalum	4.47
Mercury	4.15
Tin (grey)	3.72
Indium	3.41
Tellurium	2.38
Aluminium	1.17
Gallium	1.08
Cadmium	0.56
Titanium	0.4

Fig. 13.4 Plot of T_c against number of valence electrons (Z) for some metals



- (v) Transition metals with $R < 2$ do not exhibit superconductivity. Also, monovalent metals at ambient pressures and noble metals do not show superconductivity. Ferromagnetic metals and rare earth elements except lanthanum do not display superconductivity at ambient pressures.
- (vi) In general, superconductivity occurs in materials with large resistivity ρ . Empirically, materials with $R\rho > 10^6$ show superconductivity.

A few other observations are:

- (i) The sharpness of transition temperature depends on the state of purity of the sample. In some cases it can occur within a temperature of 0.001 K.
- (ii) Presence of traces of paramagnetic elements lowers the transition temperature or even destroys the superconductivity; for instance a few parts per million of Fe destroys superconductivity in molybdenum. One atomic percent of gadolinium lowers the transition temperature of lanthanum from 5.6 to 0.6 K.

The development of high pressure techniques has made it possible to study materials at high pressures. The earlier piston-displacement methods can be used only up to a pressure of 25 GPa. But the invention of the diamond anvil techniques has extended the pressure range to 260 GPa. Several elements have been scanned for possible superconductivity at these high pressures and, indeed many of them have revealed the existence of superconductivity. The elements which are non-superconducting at normal pressures but become superconducting at higher pressures are listed in Table 13.2 along with the pressure and T_c [11]. The onset of superconductivity in these elements could be due to different reasons. Two important mechanisms are crystal structure transition and a transition in electronic structure. Bismuth is an example of the first type. It is non-superconducting in the trigonal structure at normal pressure but becomes a superconductor with $T_c = 8.7$ K at a pressure of 9 GPa where it has a fcc structure. The onset of superconductivity in the alkaline earth metals Ca, Sr, Ba is an example of the second type; these metals undergo an s - d electronic transition at high pressure.

Table 13.2 Values of T_c for some elements which are superconducting at a higher pressure

Element	Pressure (GPa)	T_c (K)
Si	12.0	8.5
Ge	11.5	5.4
Sb	36.0	8.5
Bi	9.0	8.7
Fe	21.0	2.0
Ca	150	4.0
Sr	50.0	4.0
Ba	20.0	5.0
Cs	8.0	1.7
Ce	5.0	1.7

Table 13.3 summarizes information on elemental superconductors at ambient pressures and high pressures.

The systems covered in Tables 13.1 and 13.2 have $T_c \leq 10$ K. In view of possible applications, there has been a search for superconductors with higher T_c . A number of superconducting compounds are listed in Table 13.4. Some of them have $T_c > 10$ K. Among these Nb₃Ge has the highest T_c of 23 K (which is between liquid hydrogen and liquid nitrogen temperatures).

The subject of superconductivity is now a century old. During this period superconductivity has been observed “in most of the natural elements, alloys, thousands of inorganic compounds and even organic compounds and in the double-stranded molecule of DNA” [11]. Detailed discussion of all these systems is beyond our scope.

To complete this discussion, we mention the discovery by Bednorz and Muller [12] of superconductivity in La_{2-*x*}M_{*x*}CuO₄ (M = Ba or Sr) ceramic at $T_c \approx 40$ K. The ceramic YBa₂Cu₃O₇ was soon found to have a $T_c \approx 93$ K (which is above the liquid nitrogen temperature). We shall discuss this aspect in a later section.

13.4 Properties of Superconductors

When a metal becomes a superconductor some properties such as thermal properties, magnetic properties and thermodynamic properties show some changes. Changes in thermodynamic properties depend on whether the transition takes place with or without the presence of magnetic field. We discuss the thermal and magnetic properties in this section and the thermodynamic properties in Sect. 13.5.

Table 13.3 Summary of information on elemental superconductors

H		Ambient pressure superconductor										High pressure superconductor										He
Li		T _c (K) T _c ^{max} (K) P (GPa)										T _c ^{max} (K) P (GPa)										Ne
0.0004 14 30		Be 0.026										B 11 250										F 0.6 100
Na		Mg										Al 1.14										Cl 17.3 190
												Si 8.2 15.2										Ar
												Ge 5.35 11.5										
												Ga 1.091 7 1.4										Kr
												Zn 0.875										Br 8 150 100
												Cu										
												Ni										
												Co										
												Fe 2.1 21										
												Mn										
												Cr										
												V 5.38 16.5 120										
												Ti 0.39 3.35 56.0										
												Sc 19.6 106										
K		Ca 25 161										Y 19.5 50										O
Rb		Sr 7 50										Zr 0.546 11 30										Xe
												Nb 9.50 9.9 10										
												Mo 0.92 0.012										
												Tc 7.77 1.4										
												Ru 0.51 0.00033										
												Rh										
												Pd										
												Ag										
												Cd 0.56										
												In 3.404 5.3 11.3										
												Sn 3.722 3.9 25										
												Sb 7.193										
Cs		Ba 5 18										Pb 7.193										
1.3 1.2												Hg-α 4.1533										
Fr		Ra										Bi 8.5 9.1										Rn
		Insert Ac-Lr										Pt 2.39										
												Au										
												Ir 0.14										
												Os 0.655										
												Re 1.4										
												W 0.012										
												Ta 4.483 4.5 62										
												Hf 0.12 8.6 62										
												Ha										

Table 13.4 Values of T_c for some superconducting compounds

Compound	T_c [K]
Nb_3Ge	23.20
Nb_3Ga	20.30
Nb_3Al	18.60
Nb_3Sn	18.05
NbN	17.30
ZrN	13.70
$NiBi$	4.25
$InSb$	1.90

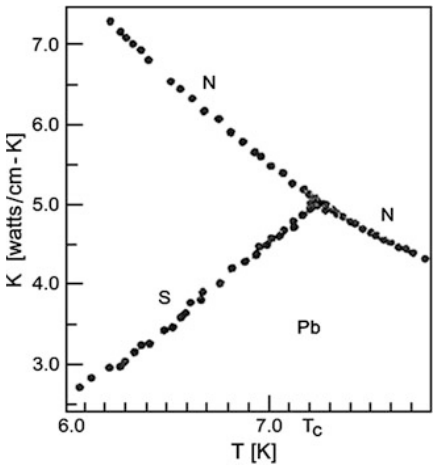
13.4.1 Thermal Properties

13.4.1.1 Thermal Conductivity

Thermal conductivity of tin is found to change drastically below the transition temperature and the value approaches zero, approximately varying as the cube of the transition temperature.

For lead, the variation in thermal conductivity with temperature is shown in Fig. 13.5. Curve N shows the thermal conductivity in the normal state and the curve S in the superconducting state. It can be seen that below T_c the thermal conductivity (K) shows a sharp decrease with decreasing temperature. On the other hand the curve for the normal state shows a continuous and smooth decrease with increasing temperature.

Fig. 13.5 Thermal conductivity of Pb as a function of temperature. Below T_c the thermal conductivity is plotted on the lower curve. The data points on the *upper* curve below T_c were obtained by application of magnetic field; they represent the normal state of Pb



Normally good electrical conductors are good heat conductors. The superconductors are good electrical conductors but are poor thermal conductors. This indicates that when a normal conductor changes to a superconductor at T_c , some fraction of electrons do not transport heat (entropy). This is the characteristic property of superconducting electrons.

13.4.1.2 Thermoelectric Effect

An electrical circuit of two superconductors or one superconductor and the other normal conductor shows none of the thermoelectric effects. The absence of Peltier effect indicates that the electrons in the superconducting phase do not carry any entropy. This shows that entropy in the superconducting state is less than the entropy in the normal state (see Sect. 13.5).

13.4.2 Magnetic Properties

13.4.2.1 Critical Field

Superconductivity may be destroyed by application of a strong magnetic field. The field may be either an externally applied field or produced by a current flowing inside the metal. The critical value of the applied field required to destroy superconductivity is denoted by H_c . The critical field H_c is a function of temperature. The variation of critical field as a function of temperature $H_c(T)$ may be represented approximately by a parabola with the following equation:

$$H_c(T) = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right] \quad (13.1)$$

where H_0 is the field necessary to destroy superconductivity at absolute zero.

The temperature variation of the critical field for a typical superconductor is shown in Fig. 13.6. At any point of the curve, the material is in the normal state above H_c and in the superconducting state below H_c . The temperature variation of H_c for some selected elements is shown in Fig. 13.7. The value of H_c for some elements is shown in Table 13.5.

13.4.2.2 Meissner Effect

Meissner and Ochsenfeld [13] discovered that when a magnetic field is applied to a superconductor at a temperature $< T_c$, the magnetic field is completely expelled (Fig. 13.8a). They measured the magnetic flux distribution outside the samples of tin

Fig. 13.6 Schematic plot of the critical magnetic field as a function of temperature

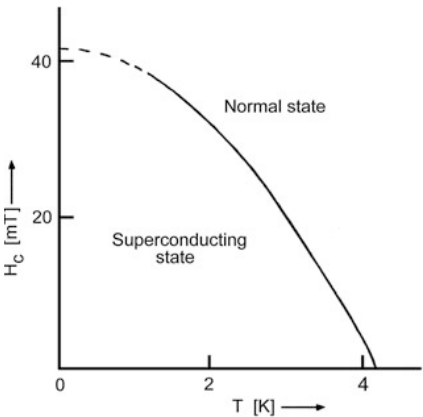


Fig. 13.7 Temperature variation of critical field for some superconducting elements

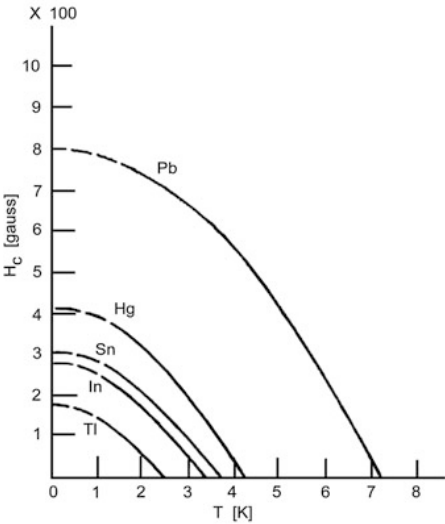


Table 13.5 Critical magnetic field (H_c) at 0 K for some elements

Element	H_c (gauss)
Nb	2060
Pb	803
Ta	830
La	1100
Zr	47
Ti	171
Ga	51
Hg	417

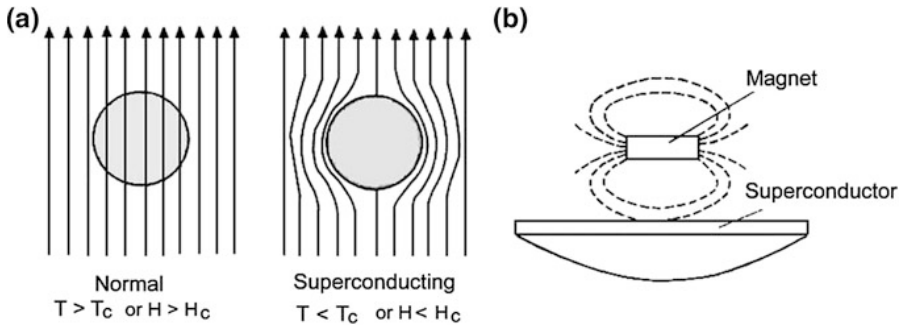
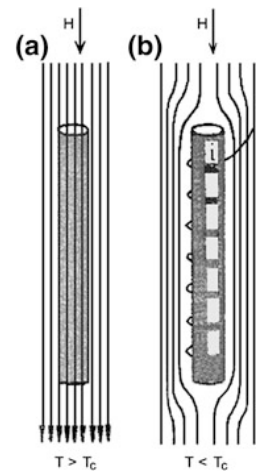


Fig. 13.8 **a** The Meissner effect and **b** principle of levitation

Fig. 13.9 Flux distribution around a superconducting rod at (a) $T > T_c$ and (b) $T < T_c$; note the flux exclusion due to induction current i



and lead placed in a magnetic field at a temperature below their transition temperature. Their experiments indicated that the magnetic behaviour for a perfect conductor and a superconductor differ. This could not be explained solely by taking into account that a superconductor is a resistanceless material. Their findings indicated that in addition to being a perfect conductor, a superconductor is also a perfect diamagnet i.e. $B/H = 0$. The applied field is completely cancelled by the induced magnetization in the specimen which opposes the applied field. This phenomenon is known as Meissner effect. The Meissner effect was first demonstrated by Arkadyev [14] by the celebrated ‘suspended magnet’ experiment (Fig. 13.8b). The magnet is lifted when the lines of force of the magnet are repelled by the surface of the superconductor. This phenomenon is known as magnetic levitation.

Consider a sample of a superconductor in the form of a long rod and apply a magnetic field parallel to the long axis as shown in Fig. 13.9a. When the sample is cooled to a temperature below T_c the flux is excluded. This is because of the induction current in the sample (Fig. 13.9b).

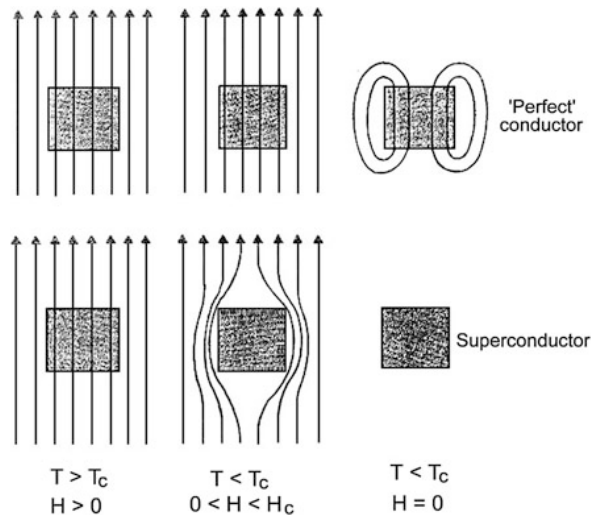


Fig. 13.10 Difference between flux distribution in a hypothetical perfect conductor and a superconductor

We now compare the flux configuration in a ‘perfect conductor’ and a superconductor in the form of solid blocks (Fig. 13.10). We shall first examine a perfect conductor (first row). Initially at $T > T_c$ and at $T < T_c$ in the presence of the field the sample acquires induced current but when the field $H = 0$ the induced field remains frozen in the sample. In the second row for the superconductor, initially it is at $T > T_c$ (normal state) in the presence of the field. When $T < T_c$ and field is below H_c , it expels the flux, since the metal is superconducting. Finally, the termination of the field induces an opposing surface current and leaves the superconductor field-free and current-free. It is to be noted that the transition from a normal to a ‘perfect conductor’ is not reversible, while in the case of a superconductor the transition between superconducting and normal state is reversible.

We now examine the flux distribution in a superconducting ring (Fig. 13.11). In the first row, when $T > T_c$ and the magnetic field is on, the flux penetrates everywhere. When the sample is cooled to $T < T_c$ and the field is still on, the flux is excluded from the cross-section of the ring by the induction current in the loop. Finally, when magnetic field is withdrawn, the flux inside the ring is trapped because the flux cannot pass through the superconductor. A large current is induced by the field which maintains the trapped flux. This current persists even after the field is withdrawn as the resistance is zero. This is called the persistent current. On the other hand, in the second row, to start with the ring is at $T > T_c$ and at zero field. It is next cooled to $T < T_c$ and the field is still zero. Finally, when the ring is at $T < T_c$ and the field is switched on, the flux is excluded from the ring as well as the hole since the flux lines are unable to cross the superconductor. Thus, in this case, in contrast to the case of the solid block, the flux configuration depends on whether the field is applied before or after the sample is cooled. In the Meissner effect, this

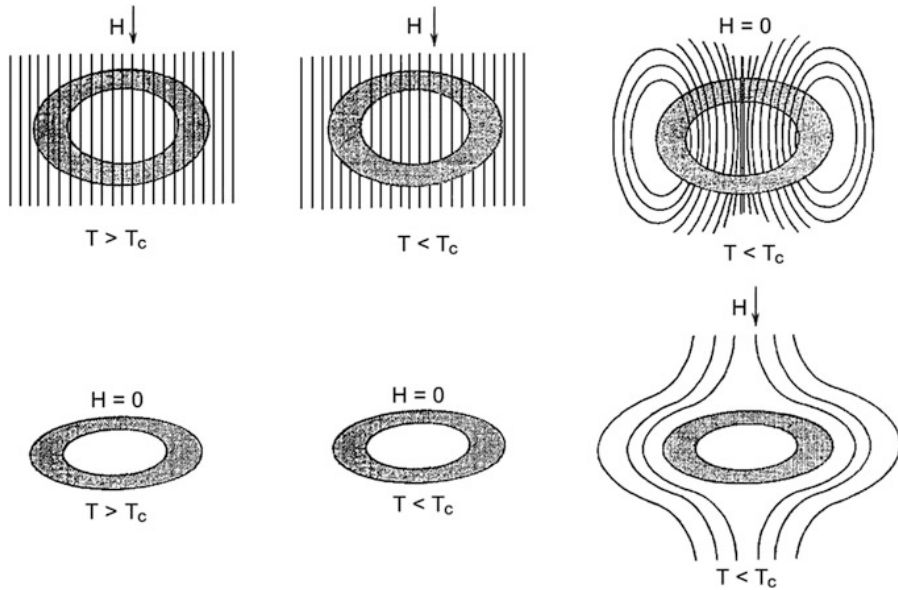


Fig. 13.11 Flux distribution around a superconducting ring with cooling and application of field in different sequences

exclusion of the magnetic flux from the interior of the sample plays an important role. Unlike the electric field which cannot penetrate a sample, a magnetic field can penetrate a superconductor within a range of 10^{-6} – 10^{-4} cm.

13.4.3 Type I and Type II Superconductors

We consider the equation for magnetic induction B in terms of the applied field H and magnetization M as

$$B = H + 4\pi M. \quad (13.2)$$

When $B = 0$, we have

$$H = -4\pi M. \quad (13.3)$$

Curves showing magnetization versus applied magnetic field are shown in Fig. 13.12. Figure 13.12a corresponds to a superconductor showing complete Meissner effect and obeying (13.2). Above the critical field H_c the specimen is a normal conductor. It is to be noted that the negative value of M represents diamagnetism. Pure specimens of many materials show this behaviour and are called type I superconductors. They are also called ‘soft superconductors’.

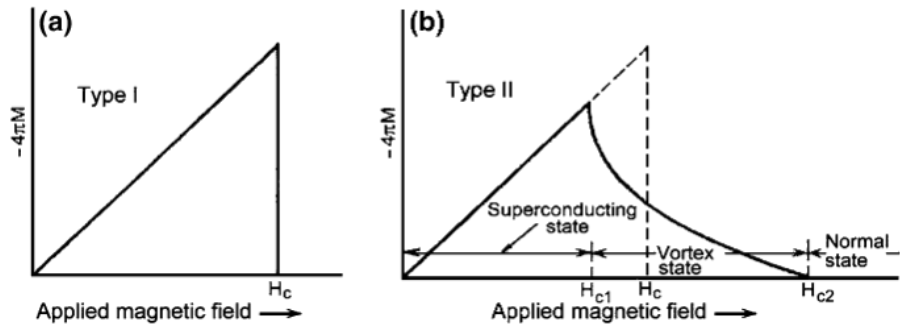


Fig. 13.12 Curves showing magnetization ($-4\pi M$) versus applied magnetic field for (a) type I and (b) type II superconductors

Table 13.6 Critical magnetic field H_{c2} at 0 K for some type II superconductors

Superconductor	H_{c2} (10^4 gauss)
Nb ₃ Ge	38
Nb ₃ Al	17.5
Nb ₃ Ga	3.4
NbN	47.0
La ₃ In	13.4
NiBi	4.2
InSb	1.9
V ₃ Si	25

There are some materials exhibiting the magnetization curve shown in Fig. 13.12b. They are termed type II superconductors or ‘hard superconductors’. While pure samples of many materials are of type I, alloys and transition metals with high resistivity values in the normal state tend to be of type II. The electrical properties of superconductors of type II differ from those of type I in that they have a vortex state in between a superconducting and a normal state.

In type II superconductors the magnetic flux is excluded completely from the sample for comparatively weak fields up to H_{c1} . Above H_{c1} in the region up to H_c there exists a partial penetration of flux. It is said to be in the vortex state. Above H_{c2} for high fields (>100 times H_c) the flux penetrates completely and the superconductivity vanishes.

Values of the critical field H_c of type I superconductors are too low for any technical application. But type II materials with high values of H_{c2} are of technical importance. Values of H_{c2} for some materials are listed in Table 13.6.

Type I may be changed to type II by addition of alloying elements. They have high electrical resistivity and short mean free path; hence they favour superconductivity. Lead is a superconductor of type I in its pure form. But with addition of 2 wt% of indium, it changes to type II. The magnetization curves with increasing

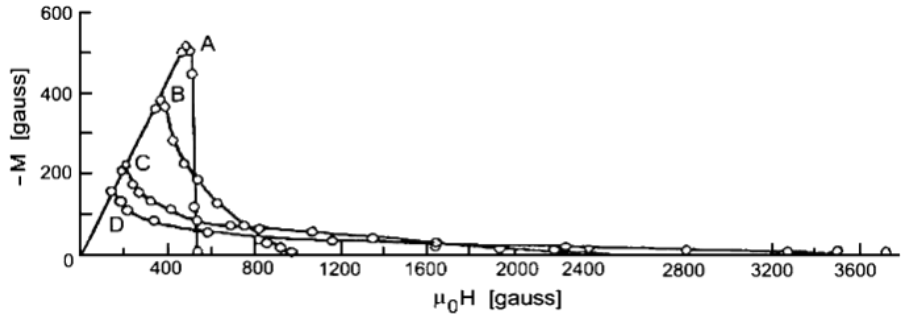
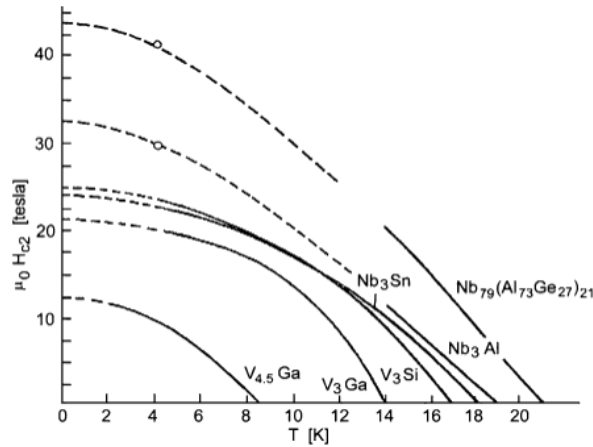


Fig. 13.13 Magnetization curves at 4.2 K for lead and lead-indium alloys: A, Pb; B, Pb-2.08 wt% In; C, Pb-8.23 wt% In; D, Pb-20.4 wt% In

Fig. 13.14 Plot of H_{c2} versus temperature for some type II superconductors



percentage of indium are shown in Fig. 13.13 [15]. It is to be noted that the areas under the curves are the same for type I (A) and type II (B, C, D) samples. Other characteristics like the transition temperature, the energy gap and the intrinsic electronic structure remain unaltered. The dependence of the critical magnetic field H_{c2} on temperature for some type II superconductors was studied by Foner et al. [16]; their results are shown in Fig. 13.14.

13.4.4 Isotope Effect

The variation of critical temperature with isotopic mass M is termed isotope effect. For instance for mercury T_c changes from 4.185 to 4.146 K for a change in the isotopic mass from 199.5 to 203.4 atomic mass units. The critical temperature of different isotopes of a metallic element varies with isotopic mass approximately as

the inverse square root of the atomic mass. This dependence of temperature on the atomic mass provides direct evidence that the ionic motion plays an important role in establishing superconductivity. The involvement of phonons in superconductivity was first proposed by Frohlich [17] and evidence in support is provided by isotope effect.

The variation of T_c with isotopic mass M may be expressed as

$$T_c = \frac{1}{M^\alpha} \quad (13.4)$$

where $\alpha \approx 0.5$. The Debye temperature θ_D also has a similar mass dependence ($\theta_{\text{Debye}} \propto M^{-1/2}$). This is another indication of the involvement of lattice vibrations in superconductivity.

It was observed in due course that $\alpha = 0.5$ is applicable to s - p metals. For some metals such as transition metal superconductors, the values range from 0 to 0.3. However the importance of isotope effect in intuitively suggesting the concept of electron-phonon interaction cannot be ignored.

13.5 Thermodynamics of Superconducting Transition

The transition from normal to superconducting state may be expressed in terms of thermodynamic quantities as in the case of transition from liquid to vapour phase. This will allow us to obtain expressions for entropy, latent heat and specific heat in terms of critical field and temperature.

We consider a type I superconductor with complete Meissner effect so that, the flux B is zero, i.e. $M = -H/4\pi$.

The Gibb's free energy G per unit volume is given by

$$G = U - TS - HM \quad (13.5)$$

where U is the internal energy, T the temperature, S the entropy, H the magnetic field and M the magnetization. In this equation, pV in the standard equation is replaced by $-HM$ as we are dealing with a magnetic system. Differentiating (13.5) and using $dU = TdS + HdM$ from the second law of thermodynamics, we have at constant temperature

$$dG = -MdH = \frac{H}{4\pi} dH. \quad (13.6)$$

For the superconducting state, the Gibb's free energy $G_s(H)$ for an increase of field from zero to H is

$$\int_0^H dG = \int_0^H \frac{H}{4\pi} dH = \frac{H^2}{8\pi} \quad (13.7)$$

or,

$$G_s(H) - G_s(0) = H^2/8\pi. \quad (13.8)$$

At $H = H_c$ the critical field, the normal and superconducting states are at equilibrium at a given temperature T . Hence

$$G_n(T, H_c) = G_s(T, H_c) = G_s(0) + H_c^2/8\pi. \quad (13.9)$$

The variation of free energy recorded for aluminium as a function of temperature is shown in Fig. 13.15a. It can be seen that for temperatures $T < T_c$, G_n is higher than G_s . The curves merge at the transition temperature.

Fig. 13.15 **a** Experimental values of G_n and G_s for Al at low temperature. G_s is measured at zero magnetic field and G_n below T_c by applying magnetic field; **b** free energies G_n and G_s versus applied magnetic field

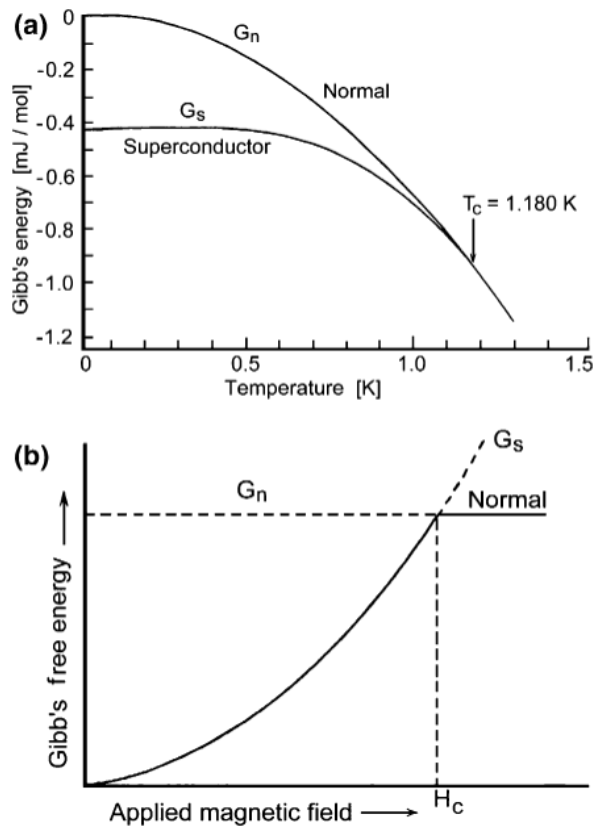
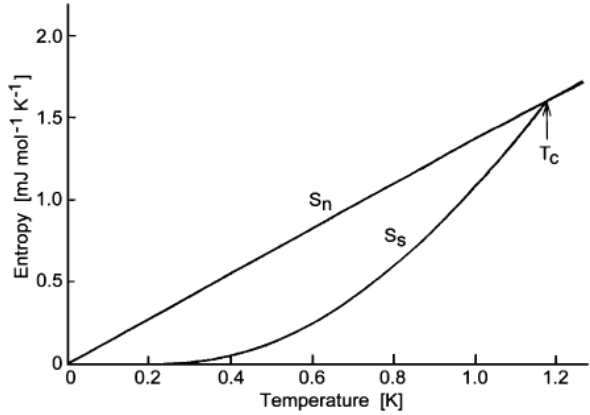


Fig. 13.16 Plot of entropy versus temperature for Al in normal (S_n) and superconducting (S_s) states



The variation of free energy for the normal and superconducting states is shown as a function of applied field in Fig. 13.15b. Up to the critical field the free energy for the superconducting phase is lower than that of the normal state. At the critical field the free energies are equal. Beyond the critical field, the free energy of the superconducting phase shows an increase. For the normal metal, the free energy is independent of the applied field.

Entropy Entropy is defined as $S = -(\partial G/\partial T)_p$. Hence, using (13.9) we get

$$\left(\frac{\partial G_n}{\partial T}\right)_p - \left(\frac{\partial G_s}{\partial T}\right)_p = -(1/8\pi)2H_c \left(\frac{\partial H_c}{\partial T}\right)_p$$

or,

$$S_n - S_s = -\frac{H_c}{4\pi} \frac{\partial H_c}{\partial T}. \quad (13.10)$$

As $(\partial H_c/\partial T)_p$ is always a negative quantity, $S_n - S_s$ is positive. In the presence of a magnetic field, the entropy in the normal state is greater than that in the superconducting state. Figure 13.16 shows the experimental results for aluminum. The lower entropy in the superconducting state shows that electrons in the superconducting state are in an orderly state.

Latent Heat The latent heat

$$Q = T(S_n - S_s) = -T \frac{H_c}{4\pi} \frac{\partial H_c}{\partial T}. \quad (13.11)$$

At $T = T_c$, $H_c = 0$. Hence the latent heat is zero at the transition temperature.

13.5.1 Specific Heat

The specific heat per unit volume C is given by $dQ/dT = TdS/dT$. Then the difference between specific heats in normal and superconducting states is given by $C_n - C_s = T \frac{\partial}{\partial T} (S_n - S_s)$. Using (13.11), we get

$$C_n - C_s = -\left(\frac{T}{4\pi}\right) \left[H_c \frac{\partial^2 H_c}{\partial T^2} + \left(\frac{\partial H_c}{\partial T} \right)^2 \right]. \quad (13.12)$$

At the transition temperature, we have $T = T_c$ and $H_c = 0$. Then (13.12) becomes

$$C_s - C_n = \Delta C = \left(\frac{T_c}{4\pi} \right) \left(\frac{\partial H_c}{\partial T} \right)^2. \quad (13.13)$$

Equation (13.13) is known as Rutger's law; it has been verified experimentally [5]. Combining (13.13) with (13.1), we get

$$C_s - C_n = \left(\frac{3}{2\pi} \right) \left(\frac{H_0^2}{T_c} \right) \left[\left(\frac{T}{T_c} \right)^3 - \frac{1}{3} \left(\frac{T}{T_c} \right) \right]. \quad (13.14)$$

Both C_s and C_n consist of two components, the lattice component (with index l) and an electronic component (with index e). We may express C_s and C_n as $C_s = C_{ls} + C_{es}$ and $C_n = C_{ln} + C_{en}$. Assuming that the lattice components are the same in the normal state and the superconducting state, we may rewrite (13.14) as

$$C_s - C_n = C_{es} - C_{en} = \left(\frac{3}{2\pi} \right) \left(\frac{H_0^2}{T_c} \right) \left[\left(\frac{T}{T_c} \right)^3 - \frac{1}{3} \left(\frac{T}{T_c} \right) \right]. \quad (13.15)$$

From the free electron theory (Chap. 3), it is known that

$$C_{en} = \gamma T. \quad (13.16)$$

Comparing (13.15) and (13.16), we get

$$\gamma = \left(\frac{1}{2\pi} \right) \left(\frac{H_0^2}{T_c^2} \right). \quad (13.17)$$

Obviously, C_{es} is the T^3 term in (13.15). Thus we have

$$C_{es} = 3\gamma T_c \left(\frac{T}{T_c} \right)^3. \quad (13.18)$$

Experimental results agree with (13.18) except at very low temperatures.

From experiments at very low temperatures on the superconducting phase, it is found that the electronic contribution C_{es} decreases exponentially. Taking this into account, (13.18) may be modified as

$$\frac{C_{es}}{\gamma T_c} = a \exp(-bT_c/T) \quad (13.19)$$

where a and b are constants. Further, (13.19) may be written as

$$\frac{C_{es}}{\gamma T_c} = a \exp(-\Delta/k_B T) \quad (13.20)$$

where Δ is called the energy gap parameter; we shall discuss this parameter in a separate section.

A second order transition is characterized by a nil latent heat at the transition point and a discontinuity in the specific heat at the transition point. From (13.11) and (13.13), we see that the superconducting transition fulfills these characteristics and, hence, is a second order transition.

We have discussed theoretically several aspects of the specific heats in the normal and superconducting states. We shall now compare these theoretical results with experimental observations:

- (i) A comparison of ΔC from (13.12) with experimental data on tin from the work of Keesom and Laer [18] is shown in Fig. 13.17. The dashed curve represents values from (13.12). It can be seen that the agreement is fair.
- (ii) Figure 13.18 shows data on specific heat of gallium in the normal and superconducting states [19]. Data for the normal state below T_c was obtained by application of a magnetic field of 200 Gauss. A sudden drop in specific heat is noticed at T_c .

Fig. 13.17 Temperature variation of ΔC for tin; smooth curve represents experimental data [18]; dashed curve from (13.12)

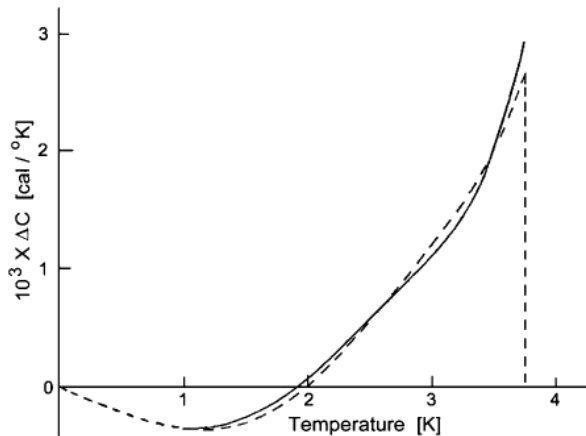


Fig. 13.18 Temperature variation of specific heat (C_p) of gallium in the normal and superconducting states [19]; values for the normal state below T_c were obtained by applying a magnetic field

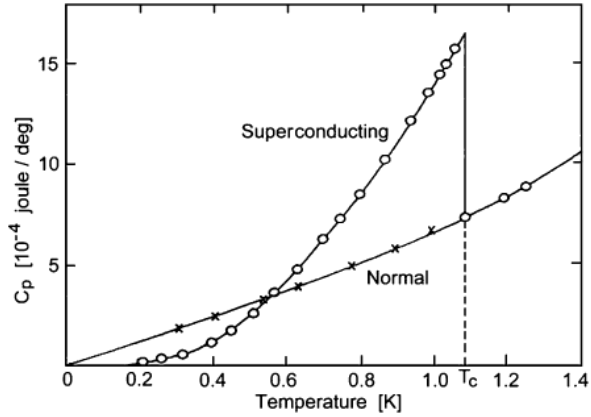
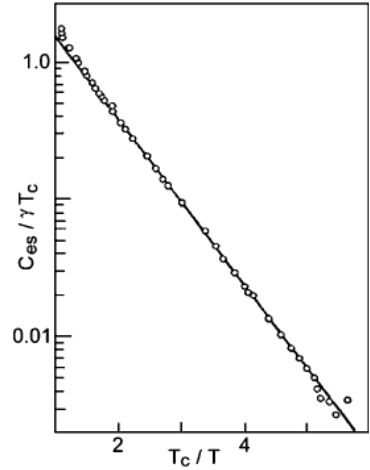


Fig. 13.19 Temperature variation of the electronic specific heat C_{es} of gallium below T_c ; the plot has slope $(\Delta/k_B T)$ in conformity with (13.20)



- (iii) A semi-log plot of the electronic specific heat C_{es} of gallium in the superconducting phase against T_c/T is shown in Fig. 13.19. The straight line plot is in conformity with (13.20).
- (iv) Figure 13.20 shows data on specific heat of NbSe_2 in the normal and superconducting phases [20]. Plot of C/T against T^2 yields γ as intercept. The figure explains how values of C_{en} , C_{es} and $\Delta C/T_c$ can be obtained. C_{ph} is the lattice specific heat which is calculated from a known value of the Debye temperature.

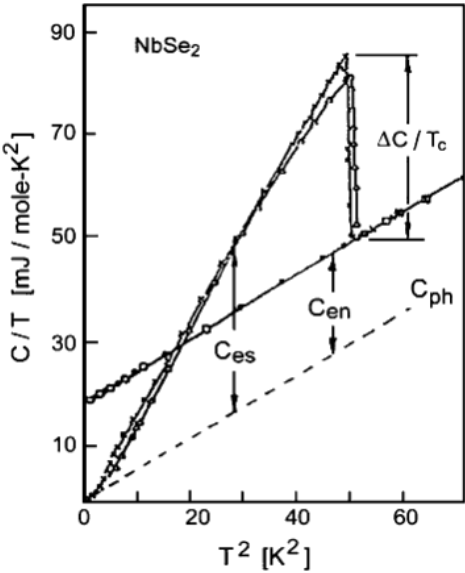


Fig. 13.20 The normal electron contribution (C_{en}), superconducting electron contribution (C_{es}) and phonon contribution (C_{ph}) to the specific heat of superconductor NbSe_2

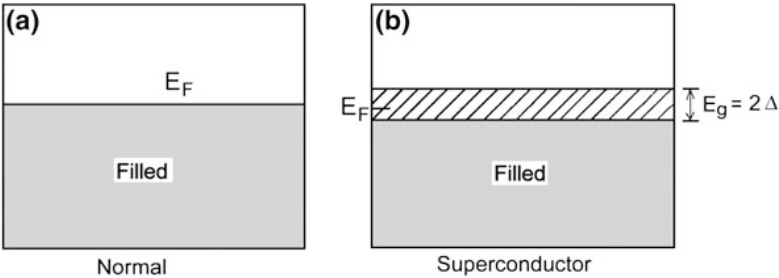


Fig. 13.21 **a** Filled conduction band in a normal metal; **b** the energy gap $E_g (=2\Delta)$ at the Fermi level E_F in a superconductor

13.5.2 Energy Gap

There is an energy gap in the energy spectrum of the electrons in a superconductor. This represents the energy required by the superconducting electrons to change over to normal electrons. The position of the energy gap (E_g) vis-à-vis the Fermi energy (E_F) and the filled band is shown in Fig. 13.21. The energy gap E_g is much smaller than the energy gap in a semiconductor. In a semiconductor the origin of the energy gap is in the electron-lattice interaction. This interaction keeps the electrons in an ordered state with respect to the Fermi gas.

Table 13.7 Values of energy gap E_g by different methods for some superconductors

Superconductor	E_g in units of $k_B T_c$		
	From specific heat	From infrared absorption	From electron tunneling
Sn	3.6	3.6	4
In	3.9	4.1	4
Pb		4.1	4.2
Hg (α)			4.6
Al			3.4
Nb			3.8
V			3.4
Zn			3.2
Cd			3.2
Ta			3.6

The energy gap in a superconductor is temperature dependent. It decreases to zero as the temperature approaches the critical temperature T_c . The energy gap E_g is twice the energy gap parameter Δ which was introduced in the expression for C_{es} (13.20).

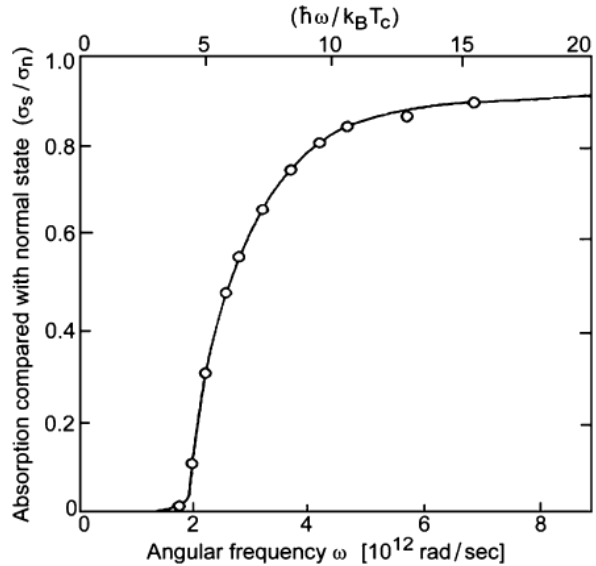
The energy gap can be determined from the experimental values of C_{es} . It can also be determined from other phenomena like radiation absorption and quantum tunneling which will be discussed later. The values of E_g for some superconducting elements are given in Table 13.7.

13.5.3 Absorption of High Frequency Electromagnetic Radiation

Superconductors have zero DC resistance at T_c . But the AC resistance is not zero though it is small. The absorption of electromagnetic radiation is studied in terms of complex resistance or complex conductivity. The absorption of electromagnetic radiation by superconductors was studied by Biondi et al. [21], Glover and Tinkham [22, 23] and by Ginsberg and Tinkham [24]. Although their experimental techniques were slightly different, the results are the same, viz., the absorption of radiation by a superconductor suddenly increases at a critical frequency ω_c which lies in the far infrared or microwave region (10^{10} – 10^{12} Hz).

Ginsberg and Tinkham made measurements of absorption by thin films in the frequency range $\omega = 10^{12}$ rad/s. Their results for absorption are given as the ratio σ_s/σ_n , σ_s and σ_n being the complex conductivities in the superconducting and normal states. They studied lead, tin and indium. Their results for indium are shown in Fig. 13.22. It can be seen that the absorption which is low increases suddenly at $\omega_c \approx 2 \times 10^{12}$ rad/s.

Fig. 13.22 Frequency variation of the absorption of electromagnetic radiation by an indium film in the superconducting state



This frequency is associated with an energy $\hbar\omega_c = 4k_B T_c$. We can interpret this absorption as selective absorption of energy by the superconducting electrons to cross over the energy gap E_g in order to become normal electrons. The optical effect is that the indium thin film which was semi-transparent at frequencies less than ω_c becomes opaque at ω_c and higher frequencies. This frequency gives the energy gap ($E_g = \hbar\omega_c$). The results obtained by Ginsberg and Tinkham are given in Table 13.7. The energy gap values obtained from absorption are in agreement with those from specific heats. It is to be noted that whereas absorption experiments give values of $E_g = 2\Delta$, the specific heat experiments give values $E_g = \Delta$.

13.6 Theories of Superconductivity

The first theoretical approach to account for superconductivity was proposed 21 years after its discovery by Gorter and Casimir [25, 26]. The theory, known as ‘the two-fluid model’, is analogous to the two-fluid theory of liquid helium. In the superconducting state the electrons consist of two types, normal electrons and superconducting electrons. We shall denote their densities or concentrations by n_n and n_s respectively. The ratio of n_n and n_s changes with temperature. The superconducting electrons do not carry any entropy and flow with zero resistance, while the normal electrons behave like the electrons in any normal metal. Above the transition temperature all the electrons are normal and as the temperature reaches 0 K, all are of the superconducting type. Thermodynamical analysis has shown that the free energy and the entropy in the superconducting phase are lesser than those of the normal phase in the temperature range $0 - T_c$ (Sect. 13.5).

The two-fluid theory could account for the experimentally observed parabolic behaviour for the critical field and T^3 law for the specific heat, but could not account for several other characteristic properties mentioned in Sect. 13.4.

In the following sections we discuss Londons' modified two-fluid theory [27] followed by an account of the microscopic theory proposed by Bardeen, Cooper and Schrieffer [28]. Finally, the theory by Ginzburg and Landau [29] will be discussed in brief.

13.6.1 London Equations

London and London [27] modified the two-fluid theory of Gorter and Casimir [25] to take into account the perfect diamagnetic property of the superconducting state. The normal and super currents are assumed to flow parallel to each other in a superconductor.

The total current density $\mathbf{J}$ is given by

$$\mathbf{J} = \mathbf{J}_s + \mathbf{J}_n \quad (13.21)$$

where $\mathbf{J}_s$ is the superconducting electron current density and $\mathbf{J}_n$ the normal current density. Since the supercurrent $\mathbf{J}_s$ flows without any resistance it is assumed to carry the entire induced current. The contribution from normal current is ignored in the present treatment.

$\mathbf{J}_s$ is the flow of current due to electrons of mass m , charge e and density n_s . Under the influence of an electric field E , the equation of motion for the super-electrons is given by

$$m \frac{d\mathbf{v}_s}{dt} = -e\mathbf{E} \quad (13.22)$$

where $\mathbf{v}_s$ is the velocity of the superelectrons. The current density is

$$\mathbf{J}_s = -n_s e \mathbf{v}_s. \quad (13.23)$$

Using (13.22) and (13.23), we get

$$\frac{d\mathbf{J}_s}{dt} = \frac{n_s e^2}{m} \mathbf{E}. \quad (13.24)$$

Equation (13.24) is Londons' first equation. In this equation if the electric field $\mathbf{E} = 0$, the current $\mathbf{J}_s$ may still be finite. In comparison, for normal current flow

given by $\mathbf{J}_n = \sigma \mathbf{E}$ (where σ is the conductivity), no current flow is possible for $\mathbf{E} = 0$. The electric field $\mathbf{E}$ and the magnetic induction $\mathbf{B}$ are related by Maxwell's relation

$$\text{Curl } \mathbf{E} = -\frac{1}{c} \frac{d\mathbf{B}}{dt} \quad (13.25)$$

where c is the velocity of light. Substituting for $\mathbf{E}$ from (13.24) in (13.25), we have

$$\text{Curl} \left[\left(\frac{mc}{n_s e^2} \right) \frac{d\mathbf{J}_s}{dt} \right] = -\frac{d\mathbf{B}}{dt}. \quad (13.26)$$

Integrating (13.26) with respect to time gives

$$\text{Curl} \left[\left(\frac{mc}{n_s e^2} \right) \mathbf{J}_s \right] = -\mathbf{B}. \quad (13.27)$$

F. London and H. London assumed that the constant of integration is zero. This also implies an assumption that in the bulk of the superconductor, where $\mathbf{B} = 0$, the current flow is irrotational. Substituting Λ for $(mc/n_s e^2)$, (13.27) may be written as

$$\text{Curl} (\Lambda \mathbf{J}_s) = -\mathbf{B}. \quad (13.28)$$

Equation (13.28) is known as Londons' second equation. If this equation is added to Maxwell's equations they describe together the superconductor characteristics. We consider Maxwell's two equations

$$\text{Curl } \mathbf{B} = (4\pi/c) \mathbf{J} \quad (13.29)$$

and

$$\text{div } \mathbf{B} = 0. \quad (13.30)$$

Using the identity,

$$\text{Curl } \text{Curl } \mathbf{B} = \text{grad } \text{div } \mathbf{B} - \nabla^2 \mathbf{B}, \quad (13.31)$$

(13.29) and (13.30) lead to

$$\nabla^2 \mathbf{B} = -\text{Curl } \text{Curl } \mathbf{B} = -\text{Curl} (4\pi/c) \mathbf{J}_s. \quad (13.32)$$

Substituting from (13.28) we get

$$\nabla^2 \mathbf{B} = \frac{4\pi}{c\Lambda} \mathbf{B}. \quad (13.33)$$

Writing $4\pi/c\Lambda = 1/\lambda^2$ we get

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda^2} \mathbf{B}. \quad (13.34)$$

where

$$\lambda = \left(\frac{mc^2}{4\pi n_s e^2} \right)^{1/2}. \quad (13.35)$$

λ is called the London penetration depth. The solution of the one-dimensional form of (13.34) is

$$B(x) = B(0)e^{-x/\lambda}. \quad (13.36)$$

Equation (13.36) shows that the solution of the type $B(x) = \text{constant}$ cannot exist. The graphical representation of (13.36) is shown in Fig. 13.23. Equation (13.36) indicates that the flux density decreases exponentially inside the superconductor falling to $1/e$ of its initial value at a distance λ , the penetration depth. Equation (13.36) shows that if $B(0)$ is the field at the plane boundary at $x = 0$, i.e. at the yz plane, the field $B(x) \rightarrow 0$ as $x \rightarrow \infty$ indicating that flux inside the bulk of the superconductor is zero. Hence (13.36) accounts for Meissner effect.

The penetration depth λ varies with temperature according to the relation

$$\lambda(T) = \lambda(0) \left(1 - \frac{T^4}{T_c^4} \right)^{-1/2}. \quad (13.37)$$

Here $\lambda(0)$ is the penetration depth at $T = 0$. $\lambda(0)$ can be estimated from (13.35) by replacing n_s with n_n . As an example, $\lambda(0) = 520 \text{ \AA}$ for Hg.

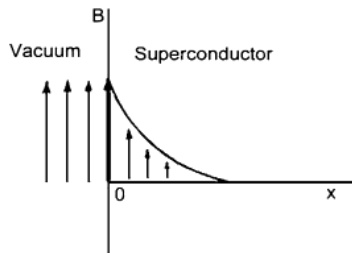


Fig. 13.23 Exponential decrease of magnetic flux (B) inside a superconductor placed in a magnetic field according to (13.36)

We now consider the effect of alternating current in a superconductor. The alternating current for the superconducting electrons is given by

$$\mathbf{J}_s = \mathbf{J}_{s0} e^{i\omega t}. \quad (13.38)$$

Using (13.24), we get

$$\mathbf{J}_s = -\frac{i n_s e^2}{\omega m} \mathbf{E}. \quad (13.39)$$

Combining (13.35) with (13.39) we get

$$\mathbf{J}_s = -\frac{i c^2}{4\pi \lambda^2 \omega} \mathbf{E}. \quad (13.40)$$

For normal electrons $\mathbf{J}_n = \sigma \mathbf{E}$, and since $\mathbf{J} = \mathbf{J}_n + \mathbf{J}_s$, we have

$$\mathbf{J} = \left(\sigma - \frac{i c^2}{4\pi \lambda^2 \omega} \right) \mathbf{E}. \quad (13.41)$$

For high frequencies, (13.41) predicts that the metal behaves like a normal metal. Hence for optical frequencies, no difference is detected between the normal and superconducting states. However, a difference is observed at lower frequencies.

According to Londons' equations, the penetration depth is a fundamental length characteristic of a superconductor. During experimental work on superconductors with high frequency field, Pippard [30] observed that λ is not a simple constant. It shows anisotropy, changes with mean free path of electrons and is temperature dependent.

13.6.2 Coherence Length

Pippard extended Londons' theory by introducing a parameter ξ characterizing the superconductor. ξ is called the 'coherence length' and is defined as the distance within which the ratio of superconducting electrons to normal electrons cannot change drastically in a spatially varying magnetic field. It is also a measure of the minimum spatial extent of a transition layer between a normal conductor and a superconductor [4].

The Londons' equation (13.28) can be expressed in terms of the vector potential $\mathbf{A}$ of the local magnetic field as

$$\text{Curl}(\Lambda \mathbf{J}_s) = -\mathbf{B} = \text{Curl} \mathbf{A}. \quad (13.42)$$

Then

$$\text{Curl } \mathbf{J}_s = -\frac{c}{4\pi\lambda^2} \text{Curl } \mathbf{A} \quad (13.43)$$

Equation (13.43) shows that the variation in $\mathbf{A}$ and $\mathbf{J}_s$ is similar. This is valid for fields which are uniform. For fields which vary rapidly in space, the average of $\mathbf{A}$ over a coherence length should be taken. This requires extra kinetic energy. We now proceed to obtain an expression for the increase in energy due to modulation following Kittel [4].

The wave equation for a plane wave is

$$\psi(x) = e^{ikx} \quad (13.44)$$

where k is the propagation vector. The wave function for a strongly modulated wave $\varphi(x)$ will be the sum of the modulated and plane waves. Thus,

$$\varphi(x) = e^{-1/2}(e^{i(k+q)x} + e^{ikx}) \quad (13.45)$$

where q is a modulating vector.

We obtain the kinetic energy using the momentum operator

$$p_k = m\dot{x} \quad (13.46)$$

and

$$\text{kinetic energy operator} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right). \quad (13.47)$$

For unmodulated wave:

The probability density is

$$\psi * \psi = e^{-ikx} e^{ikx} = 1 \quad (13.48)$$

Kinetic energy is

$$\begin{aligned} \int \psi * \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \psi dx &= \int e^{-ikx} \left(-\frac{\hbar^2}{2m} \right) \frac{\partial^2 e^{ikx}}{\partial x^2} dx \\ &= \frac{\hbar^2 k^2}{2m} \end{aligned} \quad (13.49)$$

For modulated wave:

The probability density is

$$\begin{aligned}
 \varphi * \varphi &= \frac{1}{2} \left(e^{-i(k+q)x} + e^{-ikx} \right) \left(e^{i(k+q)x} + e^{ikx} \right) \\
 &= \frac{1}{2} (2 + e^{iqx} + e^{-iqx}) \\
 &= 1 + \cos qx.
 \end{aligned} \tag{13.50}$$

Kinetic energy is

$$\begin{aligned}
 \int \varphi * \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \varphi dx &= \frac{1}{2} \left(\frac{\hbar^2}{2m} \right) [(k+q)^2 + k^2] \\
 &= \frac{\hbar^2}{2m} k^2 + \frac{\hbar^2}{2m} kq + \frac{\hbar^2}{2m} q^2.
 \end{aligned} \tag{13.51}$$

Neglecting the q^2 term, the extra energy required to modulate is given by the second term of (13.51) as $\frac{\hbar^2}{2m} kq$. If this energy exceeds the energy gap, superconductivity will be destroyed. The critical value q_0 of the modulated wave is given by

$$\left(\frac{\hbar^2}{2m} \right) k_F q_0 = E_g \tag{13.52}$$

where k_F is the value of k at the Fermi surface. The intrinsic coherence length ξ_0 is given by $1/q_0$; hence we have

$$\xi_0 = \frac{\hbar^2 k_F}{2mE_g}. \tag{13.53}$$

In terms of the electron velocity at the Fermi surface v_F , where $v_F = \frac{\hbar k_F}{m}$, we have

$$\xi_0 = \frac{\hbar v_F}{2E_g} = \frac{\hbar v_F}{k_B T_c}. \tag{13.54}$$

A similar result was obtained by Faber and Pippard [31]; ξ_0 is given by

$$\xi_0 = a \frac{\hbar v_F}{k_B T_c} \tag{13.55}$$

where a is a constant obtained empirically. BCS theory [28] gives

$$\xi_0 = 0.18 \frac{\hbar v_F}{k_B T_c}. \tag{13.56}$$

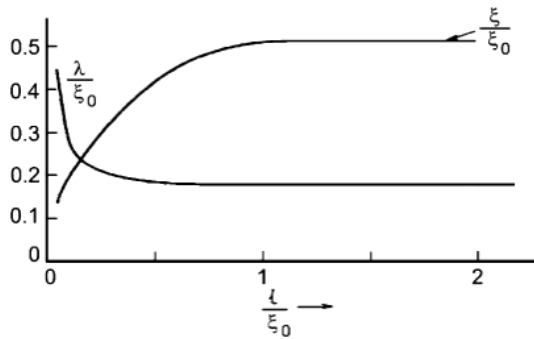


Fig. 13.24 Variation of the coherence length ζ and the penetration depth λ with the mean free path l (all parameters are expressed as fractions of the intrinsic coherence length ξ_0)

The coherence length ζ and the penetration depth λ are functions of mean free path l of conduction electrons in the normal state. The coherence length ζ is related to the intrinsic coherence length ξ_0 and the mean free path l as

$$\frac{1}{\zeta} = \frac{1}{\xi_0} + \frac{1}{l}. \quad (13.57)$$

These functions are shown in Fig. 13.24.

13.6.3 BCS Theory of Superconductivity

Nearly half a century after the discovery of superconductivity, Bardeen, Cooper and Schreiffer [28] proposed a comprehensive, microscopic theory (since popularly known as the BCS theory). The theory involves advanced mathematical techniques beyond our scope. Hence a qualitative treatment is provided.

Frohlich [17] and Bardeen [32] suggested independently that absorption and emission of phonons by electrons in a phonon field could result in an attractive interaction responsible for superconductivity. Frohlich, while dealing with work concerning polaron states in ionic crystals, suggested that the electron-phonon interaction plays a role in superconductivity. Further he suggested that if electron-phonon interactions are important, T_c should be proportional to the Debye temperature θ_D for isotopes of a given element. As θ_D is known to be proportional to $M^{-1/2}$ where M is the isotopic mass, T_c should also vary as $M^{-1/2}$. Subsequent experimental results on T_c of isotopes of several elements indeed confirmed the $T_c - M^{-1/2}$ relation and thus lent support to Frohlich's suggestion of the important role of electron-phonon interaction in superconductivity.

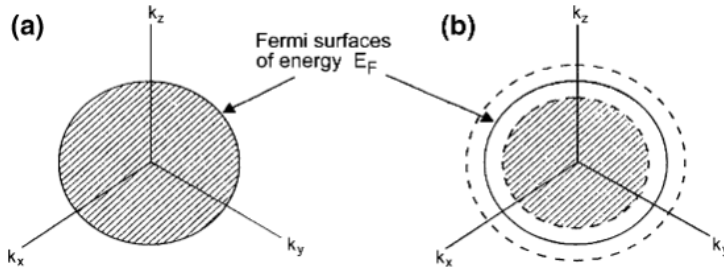


Fig. 13.25 **a** Fermi level in a normal metal with all states below E_F being occupied; **b** the BCS ground state

According to London [33], since superconductivity is manifested as a quantum phenomenon like superfluidity, the magnetic flux should be quantized in units of $2\pi\hbar c/e$. Subsequent experiments by Deaver and Fairbank [34] showed that the flux is indeed quantized, but the quantized unit of flux has half the value i.e. $\pi\hbar c/e$. This implies that the superconducting carriers have a charge $-2e$ which agrees with the concept of Cooper pairs. Cooper [35] proposed that states involving pairs of electrons with attractive interaction are possible. Further, the two electrons moving in a field of phonons could interact mutually by exchange of phonons called virtual phonons. A virtual phonon is one which according to the uncertainty principle does not last long enough to be observed before it is absorbed. These electrons are called Cooper pairs. It is important to note that the BCS theory does not use a boson condensation process to explain superconductivity. The electron pairs form the superconducting carriers and unpaired electrons form the normal carriers.

The effect of attractive interaction on all the electrons in the superconductor at zero temperature is that they are described by the BCS ground state rather than the occupied Fermi level of the normal state (Fig. 13.25a, b).

The BCS ground state is made up of Cooper pairs in which if one electron state is unoccupied, so is the other. It differs from the normal state as some electron states just outside the normal Fermi surface are occupied, some just inside are unoccupied (Fig. 13.26). In a BCS state, the orbitals are occupied in pairs. If an orbital with wave vector $\mathbf{k}$ and spin up $\uparrow$ is occupied, then the orbital with wave vector $-\mathbf{k}$ and spin down $\downarrow$ is also occupied. Also if $\mathbf{k}_1\uparrow$ is vacant, $-\mathbf{k}_1\downarrow$ is also vacant. The Cooper pairs have zero spin and have many characteristics of bosons. The electron-phonon-electron interaction, which could make the coulombic repulsion between the two electrons smaller, favours superconductivity (Fig. 13.27). A Cooper pair ($\mathbf{k}_1\uparrow-\mathbf{k}_1\downarrow$) can go to any other unoccupied position of a pair ($\mathbf{k}_2\uparrow-\mathbf{k}_2\downarrow$) by the exchange of a virtual phonon q .

The Cooper pair is a bound state. The energy difference between the paired state and the unpaired state is the energy gap E_g ; E_g is twice the 'energy gap parameter Δ ' i.e. $E_g = 2\Delta$. An energy E_g is required to break a Cooper pair resulting in the

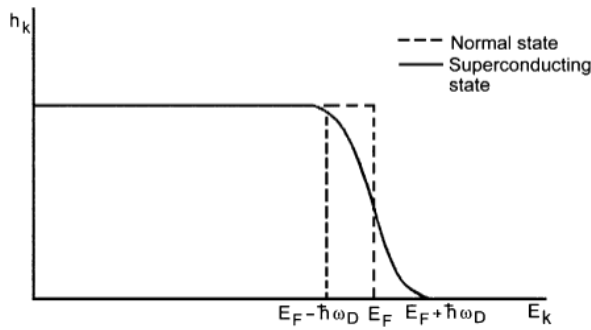
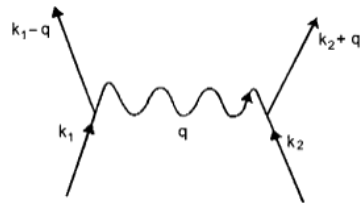


Fig. 13.26 Occupation probability h_k of electrons in a superconducting BCS state; some electrons from the normal state in energy range $(E_F - \hbar\omega_D)$ to E_F are now occupying energy states in the range E_F to $(E_F + \hbar\omega_D)$

Fig. 13.27 Exchange of virtual phonons between electrons



creation of two quasi-particle excitations. The BCS theory gives the following relation for 2Δ at 0 K.

$$\frac{2\Delta(0)}{k_B T_c} = 3.52. \quad (13.58)$$

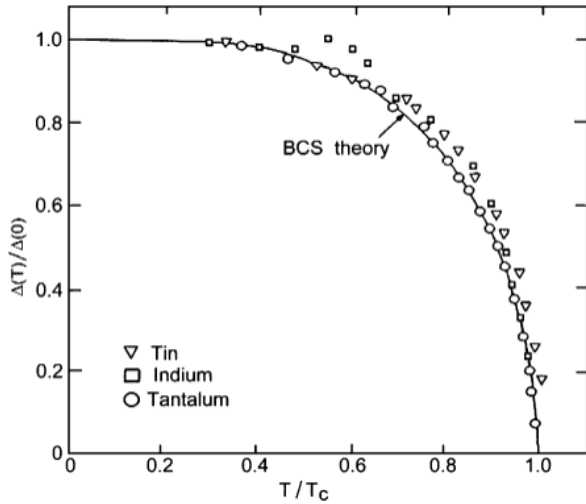
This is a universal relation in the sense that it does not contain any phenomenological quantities. Values of E_g determined by experimental methods for several elements are given in Table 13.7. Most of them are consistent with (13.58) although in the case of Pb and Hg there is significant deviation.

The BCS theory predicts a temperature dependent energy gap. In the vicinity of T_c , the temperature variation is given by

$$\frac{\Delta(T)}{\Delta_0} = 1.76 \left(1 - \frac{T}{T_c} \right)^{1/2}. \quad (13.59)$$

It follows from (13.59) that $\Delta(T) \rightarrow 0$ as $T \rightarrow T_c$. Another interpretation of (13.59) would be that the fraction of superconducting electrons decreases with increasing temperature until at $T = T_c$, all electrons are normal electrons. Experimental results on temperature variation of $\Delta(T)$ for some superconductors are shown in Fig. 13.28. These results are consistent with the BCS theory (13.59).

Fig. 13.28 Temperature dependence of the superconducting energy gap; experimental results are compared with the BCS theory



The most important concept introduced in the BCS theory is the concept of a bound state of two electrons—the Cooper pairs. From this concept several observed properties of superconductors like the critical field, specific heat, spectroscopic absorption, Meissner effect and even the London equation follows in a natural way.

13.6.4 Ginzburg–Landau (GL) Theory

Ginzburg and Landau [29] proposed a theory for superconductivity which is related to Landau's theory of second order phase transitions. They describe the superconducting state in terms of an order parameter $\psi(r)$ which vanishes above T_c and whose magnitude measures the degree of superconducting order at any position. ψ is interpreted as the wave function of the condensed super electrons and is related to the local density of superconducting electrons n_s as

$$n_s = |\psi(r)|^2. \quad (13.60)$$

GL theory in full form is too complicated to present here. Instead we consider the theory in a phenomenological way to highlight the important assumptions and results.

- (i) The phase transition is characterized by the order parameter which is identified with the macroscopic wave function ψ . ψ is a function which varies with space.
- (ii) The order parameter vanishes above T_c and its magnitude measures the degree of superconducting order below T_c .

- (iii) The order parameter can be viewed as a one-particle wave function describing the position of centre of mass of an electron pair. A single wave function is assumed as all the electron pairs are in the same two-electron state.
- (iv) The free energy is written as a function of ψ and vector potential $\mathbf{A}$. By minimizing the free energy, ψ and the supercurrent are obtained.

The theory accounts for all the parameters of a superconductor: penetration depth, coherence length and flux quantization.

The basic postulate of the GL theory is that if ψ is small and varies slowly in space, the free energy density f for a particle can be expanded in a series as a function of ψ (where $\psi = |\psi|e^{i\phi}$) as

$$f = f_n + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m^*} \left| \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right) \psi \right|^2 + \frac{h^2}{8\pi}. \quad (13.61)$$

Here, f_n is the free energy density in the normal phase, α and β phenomenological parameters, $h^2/8\pi$ the magnetic energy density, m^* and e^* are the mass and charge of the particle.

The following differential equations for ψ and the current J are derived using variational principles:

$$\alpha\psi + \beta|\psi|^2\psi + \frac{1}{2m^*} \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right)^2 \psi = 0 \quad (13.62)$$

and

$$\mathbf{J} = \frac{e^*}{m^*} \left(\hbar \nabla \phi - \frac{e^*}{c} \mathbf{A} \right) |\psi|^2. \quad (13.63)$$

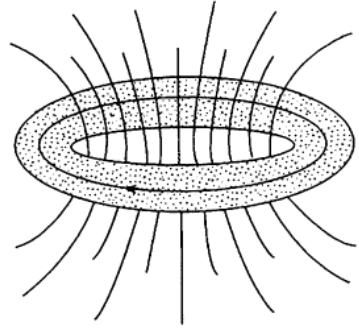
Equations (13.62) and (13.63) are referred to as GL differential equations. The quantities e^* and m^* in (13.61–13.63) are likely to be construed to have normal electronic values. However, assuming that the ‘particle’ is an electron pair, we consider $m^* = 2m$ and $e^* = 2e$ [7].

The first equation, (13.62) has a form similar to Schrödinger equation for a particle of charge $2e$ and mass $2m$ and an energy eigen value $-\alpha$. GL theory is a generalization of Londons’ theory to deal with the variation of n_s as n_s is equal to $|\psi(r)|^2$.

13.6.4.1 Flux Quantization

The total magnetic flux that passes through a superconducting material in the form of a ring assumes only quantized values.

Fig. 13.29 Magnetic flux lines about a ring-shaped superconductor



Let us consider a superconducting material in the shape of a ring as in Fig. 13.29. Since the magnetic field cannot penetrate through the material, the flux lines will be enclosed within the ring.

The current inside the ring takes a closed path. Hence, using (13.63) for current density, we have

$$\oint \mathbf{J} \cdot d\mathbf{l} = \oint \left(\hbar \nabla \phi - \frac{e^*}{c} \mathbf{A} \right) \cdot d\mathbf{l} = 0. \quad (13.64)$$

Using Stoke's theorem,

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \nabla \times \mathbf{A} \cdot d\mathbf{s} = \int \mathbf{B} \cdot d\mathbf{s} = \Phi. \quad (13.65)$$

where Φ is the flux enclosed in the ring. Since the order parameter $\psi(r)$ is single valued, its phase change around the close path must be an integral multiple of 2π . Thus

$$\oint \nabla \phi \cdot d\mathbf{l} = 2\pi n. \quad (13.66)$$

Combining (13.64), (13.65) and (13.66) we get

$$\Phi = \frac{nhc}{2e} = n\Phi_0. \quad (13.67)$$

where $\Phi_0 = hc/2e$. Φ_0 is known as a fluxoid or flux quantum. On substitution of numerical values, $\Phi_0 = 2.067 \times 10^{-7}$ gauss-cm².

13.6.4.2 Type II Superconductors

In Sect. 13.4.3 we have discussed some of the characteristics of type II superconductors. An important difference between type I and type II superconductors is in the mean free path of conduction electrons in the normal state. As mentioned earlier, the mean free path changes on alloying. Figure 13.24 shows the penetration depth λ and the coherence length ξ as a function of the mean free path l . If $\xi < \lambda$, the superconductivity will be of type II.

The ratio λ/ξ is termed Ginzburg–Landau parameter κ . Abrikosov [36] classified the superconductors in terms of the GL parameter. For pure metals which are of type I, $\kappa < 1/\sqrt{2}$ and for type II superconductors $\kappa > 1/\sqrt{2}$. The magnetization curves for some superconductors with the same H_c and different values of κ are shown in Fig. 13.30. The curve for $\kappa < 1/\sqrt{2}$ is of type I and those for $\kappa \sim 0.8$ and $\kappa = 2$ are of type II. The area under the curve is given by $H_c^2/8\pi$ for all the cases.

The magnetic phase diagram for a type II superconductor is shown in Fig. 13.31. Below H_{c1} all the flux is expelled as it is the Meissner state. Between H_{c1} and H_{c2} it is the vortex state. In the vortex state, the vortices act as tiny magnets and repel each other. The superconducting currents will be in the form of vortices within the bulk

Fig. 13.30 Magnetization curves for different values of the G–L parameter κ

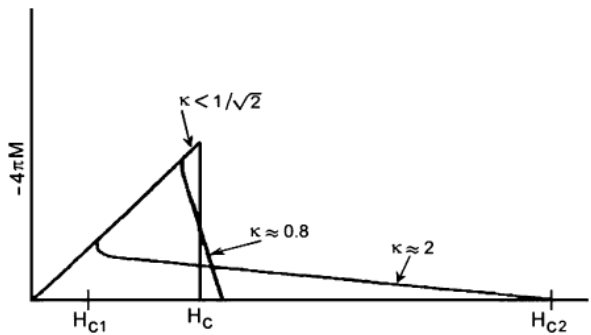


Fig. 13.31 Magnetic phase diagram for a type II superconductor

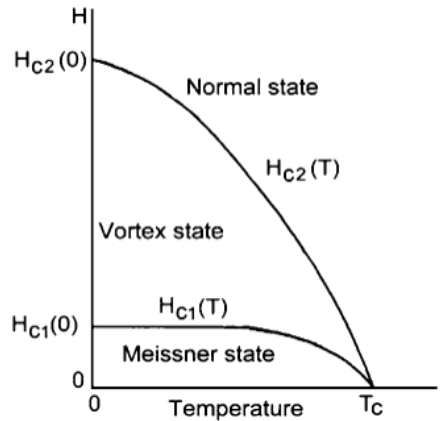


Fig. 13.32 Schematic diagram of a triangular vortex array

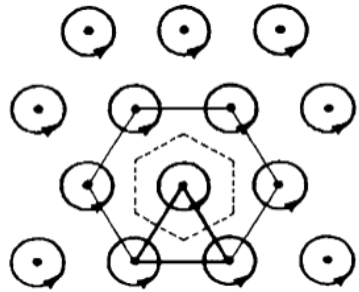
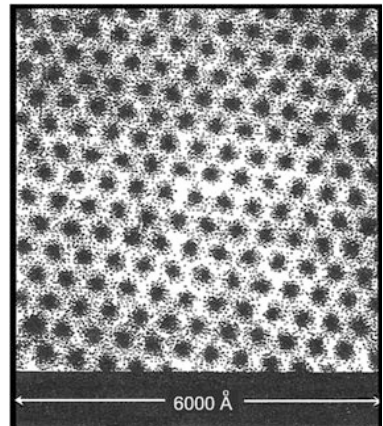


Fig. 13.33 Scanning electron micrograph of a vortex pattern for NbSe_2



of the specimen. In the vortex state, the flux penetrates in the form of quantized flux $\Phi_0 = hc/2e$. The vortex state is stable when the penetration of the applied field causes the surface energy to become negative. The regions of high density states, ‘the vortices’, form symmetric arrays. The schematic picture of such an array is shown in Fig. 13.32. A scanning electron micrograph of a typical type II superconducting material reproduced from Abrikosov [37] is shown in Fig. 13.33. The black star-like regions are images of vortices.

13.7 Normal and Josephson Tunneling

Tunneling of electrons through thin insulating barriers between metals and superconductors was investigated for the first time by Giaever [38]. Josephson [39] showed that tunneling of electron pairs is also possible with very thin insulator layers between two superconductors; his work led to the startling result that in suitable junctions, an applied voltage V results in an AC current with frequency $(2eV/\hbar)$.

In this section, we shall discuss both normal and Josephson tunneling followed by the principle and working of a SQUID.

13.7.1 Normal Tunneling

When two normal metals are separated by an insulator, the insulator acts as a barrier for the conduction of electrons. If the barrier is a thin oxide layer of the order of 10–20 Å, there is a finite probability of the electrons to pass through the barrier. Schematic diagrams of energy versus density of states are shown in Fig. 13.34. Figure 13.34a shows two normal metals n, n with an insulator i at $V = 0$. Both the metals have the same Fermi level E_F . In this situation the current is zero. If a potential difference V is applied across the metals, it results in the shift of energy levels as shown in Fig. 13.34b. Tunneling also takes place with more electrons traversing from one metal to the other. The plot between current (I_{nn}) and voltage (V) is shown in Fig. 13.34c. The variation obeys Ohm's law.

Let us now consider a combination of a normal metal n and a superconductor s with an insulating layer i between. This device s–i–n shown in Fig. 13.35a is fabricated by depositing first a thin film of superconductor on a smooth glass surface. The surface is coated by an aluminium layer which is oxidized to form the insulator. A second coating of a normal metal is then applied.

For depicting the energy levels, we shall adopt a semiconductor model for the superconductor i.e. we show the energy gap $E_g = 2\Delta$ as if it is the gap between two bands (Fig. 13.35a). Initially, the Fermi level E_F of the metal is at the middle of the

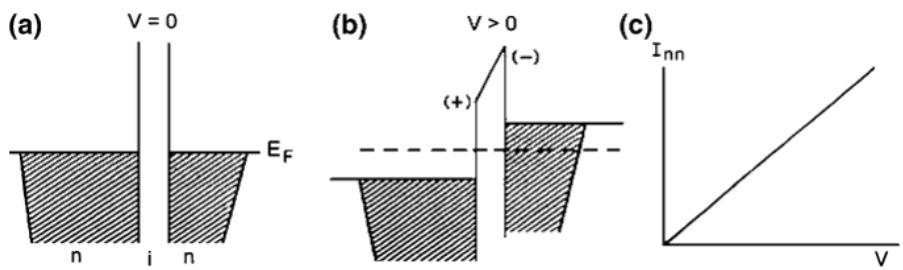


Fig. 13.34 Energy levels of electrons on the two sides of a metal-insulator- metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{nn} versus V

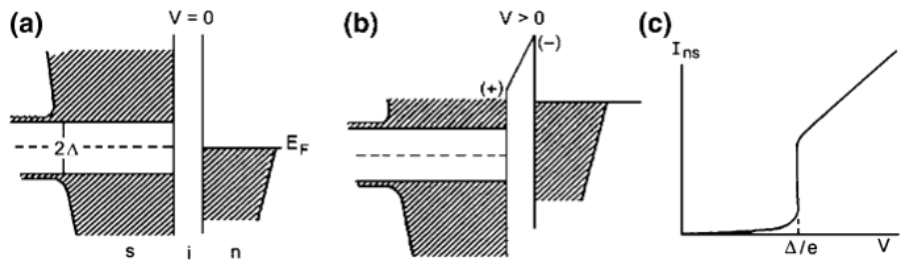


Fig. 13.35 Energy levels of electrons on the two sides of a superconductor- insulator- metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{ns} versus V

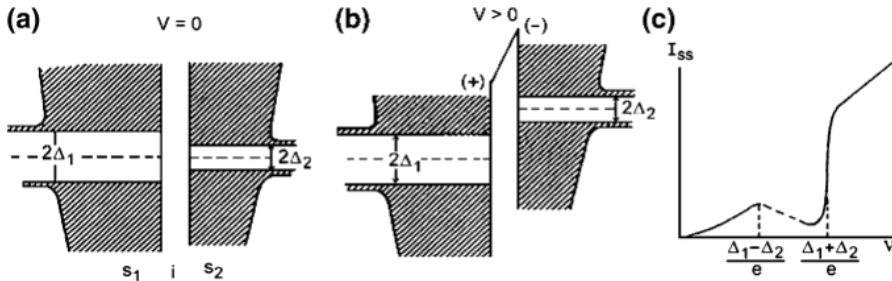


Fig. 13.36 Energy levels of electrons on the two sides of a superconductor-insulator-superconductor metal junction: **a** at $V = 0$, **b** at $V > 0$ and **c** plot of I_{ss} versus V

energy gap of the superconductor. The shaded regions indicate occupied states. When a voltage is applied, the Fermi level of the metal is shifted upwards (higher) with respect to the middle of the energy gap of s (Fig. 13.35b). However, since s is a superconductor, current cannot pass until the voltage V is equal to Δ/e or $E_g/2e$. The I_{ns} versus V plot for this device is shown in Fig. 13.35c. The rise of the current at V is sharp at $T = 0$; at higher temperatures, the onset of the current is less sharp and there is a small current even at $V < (\Delta/e)$; this is due to the thermally excited electrons. The important point is that this provides a simple method for the determination of the energy gap E_g .

We now consider tunneling between two superconductors having different energy gaps $2\Delta_1$ and $2\Delta_2$. The experiments by Giaever [38] were done with aluminium and lead with aluminium oxide barrier. Figure 13.36a shows the initial situation where s_1, s_2 correspond to lead and aluminium respectively. They have their centres of energy gap at the same level. A potential difference is applied to raise the energy levels of Al with respect to those of Pb (Fig. 13.36b). This encourages tunneling from Al to Pb (i.e. s_2 to s_1), resulting in current I_{ss} . The I_{ss} - V curve is shown in Fig. 13.36c. As the top of the s_2 -gap passes the top of the s_1 -gap a peak is observed at a voltage $V_1 = (\Delta_1 - \Delta_2)/e$. Thereafter, the current decreases as the number of available states decreases. Again when the bottom of the energy gap of s_2 passes the top of s_1 -gap there is an increase in current corresponding to a voltage $V_2 = (\Delta_1 + \Delta_2)/e$. From the voltages V_1 and V_2 , the energy gaps of both the superconductors are obtained.

In the above cases, it is essentially a single electron that tunnels through the barrier. Hence the process is called ‘single particle tunneling’.

13.7.2 Josephson Tunneling

We have discussed single particle tunneling where the oxide barrier is 10–20 Å thick. Josephson [39] predicted that in addition to the ‘normal tunneling’ of single electrons, supercurrents consisting of pairs of electrons can flow across a barrier. Such an effect

was experimentally observed and is called ‘quantum tunneling’. Physically it means that the Cooper pairs might tunnel under suitable conditions with zero potential difference across very thin barriers; the bound electrons traverse through the junction from one superconductor to the other without dissociating. In effect, a significant overlap of the phases occurs to establish a weak link between the two superconductors. This results in a current flow and is referred to as DC Josephson effect.

In normal tunneling, the barrier is 10–20 Å thick; it is made of Al_2O_3 . Josephson effect requires thinner barriers. Al_2O_3 is not suitable for making such thin barriers. Instead oxides of Sn and Pb in the form of films are employed [40].

Josephson also predicted that if a DC voltage is applied across the junctions, it causes an alternating current flow. This is referred to as AC Josephson effect.

13.7.2.1 DC Josephson Effect

Let ψ_1 be the probability amplitude of electron pairs on one side of a junction and ψ_2 on the other. Initially we assume that the two superconductors are at the same potential.

The time dependent Schrödinger equations applied to ψ_1 and ψ_2 are

$$i\hbar \frac{\partial \psi_1}{\partial t} = \hbar k_0 \psi_2 \text{ and } i\hbar \frac{\partial \psi_2}{\partial t} = \hbar k_0 \psi_1 \quad (13.68)$$

where $\hbar k_0$ represents electron pair coupling interaction across the insulator. If the insulator is very thick $k_0 = 0$ and pair tunneling cannot take place.

Let us assume the solution to be of the form

$$\psi_1 = n_1^{1/2} e^{i\phi_1} \text{ and } \psi_2 = n_2^{1/2} e^{i\phi_2} \quad (13.69)$$

where n_1 and n_2 are the Cooper pair concentrations in the two superconductors and ϕ_1 and ϕ_2 are the phases. From (13.68) in (13.69) we get

$$\frac{\partial \psi_1}{\partial t} = \frac{1}{2} n_1^{-1/2} e^{i\phi_1} \frac{\partial n_1}{\partial t} + i\psi_1 \frac{\partial \phi_1}{\partial t} = -ik_0 \psi_2 \quad (13.70)$$

and

$$\frac{\partial \psi_2}{\partial t} = \frac{1}{2} n_2^{-1/2} e^{i\phi_2} \frac{\partial n_2}{\partial t} + i\psi_2 \frac{\partial \phi_2}{\partial t} = -ik_0 \psi_1 \quad (13.71)$$

Multiplying (13.70) by $n_1^{1/2} e^{-i\phi_1}$ and (13.71) by $n_2^{1/2} e^{-i\phi_2}$, we get

$$\frac{1}{2} \frac{\partial n_1}{\partial t} + in_1 \frac{\partial \phi_1}{\partial t} = -ik_0 n_1^{1/2} n_2^{1/2} e^{i\delta} \quad (13.72)$$

and

$$\frac{1}{2} \frac{\partial n_2}{\partial t} + in_2 \frac{\partial \phi_2}{\partial t} = -ik_0 n_1^{1/2} n_2^{1/2} e^{-i\delta} \quad (13.73)$$

where

$$(\phi_2 - \phi_1) = \delta. \quad (13.74)$$

Equating the real and imaginary parts in (13.72) and (13.73) gives

$$\frac{\partial n_1}{\partial t} = 2k_0(n_1 n_2)^{1/2} \sin \delta; \quad \frac{\partial n_2}{\partial t} = -2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.75)$$

and

$$\frac{\partial \phi_1}{\partial t} = -k_0 \left(\frac{n_2}{n_1} \right)^{1/2} \cos \delta; \quad \frac{\partial \phi_2}{\partial t} = -k_0 \left(\frac{n_1}{n_2} \right)^{1/2} \cos \delta. \quad (13.76)$$

Since $n_1 \approx n_2$, we have from (13.76)

$$\frac{\partial \phi_1}{\partial t} = \frac{\partial \phi_2}{\partial t}$$

or

$$\frac{\partial}{\partial t} (\phi_1 - \phi_2) = 0 \quad (13.77)$$

Further, from (13.75) we find that

$$\frac{\partial n_2}{\partial t} = -\frac{\partial n_1}{\partial t} \quad (13.78)$$

The current flow J from one superconductor to the other through the junction is proportional to $\partial n_2 / \partial t$ or $\partial n_1 / \partial t$. Equations (13.75) and (13.76) show that the current across the junction depends on the phase difference δ .

$$J = J_0 \sin \delta = J_0 \sin(\phi_2 - \phi_1). \quad (13.79)$$

The result shows that

- (i) J_0 is proportional to k_0 and represents the maximum zero voltage current that can be passed through the junction.
- (ii) with zero applied voltage a DC current will flow across the junction, with values between J_0 and $-J_0$ depending on the phase difference $\phi_2 - \phi_1$.

13.7.2.2 AC Josephson Effect

Let us now consider the case when a DC voltage V_0 is applied across the junction i.e. the insulator. An electron pair will be subjected to a potential energy difference $2eV_0$. This means that, on one side the pair is at a potential energy $-eV_0$ and on the other side the potential energy is $+eV_0$.

The equation of motion is given by

$$i\hbar \frac{\partial \psi_1}{\partial t} = \hbar k_0 \psi_2 - eV_0 \psi_1; \quad i\hbar \frac{\partial \psi_2}{\partial t} = \hbar k_0 \psi_1 - eV_0 \psi_2. \quad (13.80)$$

Proceeding in the same manner as in the case of DC Josephson effect, we obtain the following equations for $\partial n / \partial t$ and $\partial \phi / \partial t$.

$$\frac{\partial n_1}{\partial t} = 2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.81)$$

and

$$\frac{\partial \phi_1}{\partial t} = \frac{eV_0}{\hbar} - k_0(n_2/n_1)^{1/2} \cos \delta. \quad (13.82)$$

Similarly,

$$\frac{\partial n_2}{\partial t} = -2k_0(n_1 n_2)^{1/2} \sin \delta \quad (13.83)$$

and

$$\frac{\partial \phi_2}{\partial t} = -\frac{eV_0}{\hbar} - k_0(n_1/n_2)^{1/2} \cos \delta. \quad (13.84)$$

Assuming $n_1 \approx n_2$, we get from (13.82) and (13.84)

$$\frac{\partial}{\partial t}(\phi_2 - \phi_1) = \frac{\partial \delta}{\partial t} = -\frac{2eV_0}{\hbar}. \quad (13.85)$$

On integration (13.85) gives

$$\delta(t) = \delta(0) - \frac{2eV_0 t}{\hbar} \quad (13.86)$$

The superconducting current is given by

$$J = J_0 \sin[\delta(0) - 2eV_0 t / \hbar] \quad (13.87)$$

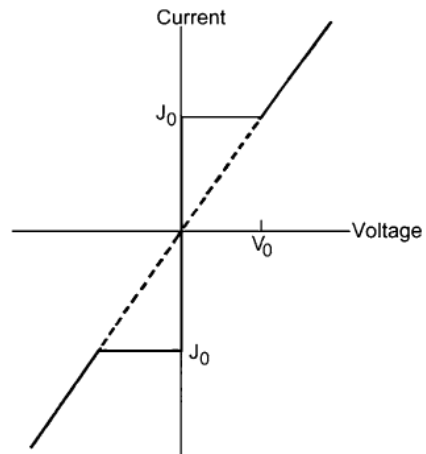
Equation (13.87) shows that the current oscillates with a frequency $\omega = 2eV_0/\hbar$. Thus if two superconducting elements are at different electric potentials, the transfer of a pair of electrons from one side to the other involves an energy change of eV_0 . This appears as a photon of energy $\hbar\omega = 2eV_0$.

The tunnel current arises from a transition between a state in which a pair is at one phase on one side and at a different phase on the other side. The relative phase of the two states oscillates at the beat frequency corresponding to the energy difference $\hbar\omega = 2eV_0$. This gives a current that oscillates at that frequency as in the case of atomic transitions.

A photon of energy $\hbar\omega = 2eV_0$ is emitted or absorbed whenever an electron pair crosses the barrier. Hence AC Josephson effect permits an accurate determination of $e/\hbar$ from the relationship between voltage and the frequency (13.87). It has been observed that a DC voltage of 1 mV across a junction produces a frequency of 483.6 GHz.

We shall summarize the behaviour of a Josephson junction with reference to the current-voltage diagram (Fig. 13.37). At zero voltage, a DC current of the order of mA passes through the junction with a value between $+J_0$ and $-J_0$ depending upon the phase difference ($\phi_2 - \phi_1$) according to (13.79). If now a small DC voltage (of the order of mV) is applied, it still results in a DC current between $-J_0$ and J_0 and the phase difference gets adjusted to suit (13.79). But no potential difference appears across the junction. This process which is called the DC Josephson effect continues as the voltage is increased up to a critical value V_0 ; the corresponding value of the DC current is now J_0 . At this voltage, suddenly a potential drop shows up across the junction. For DC voltages greater than V_0 , say V_1 , an AC current starts flowing through the junction with a frequency $\omega = 2eV_1/\hbar$ according to (13.87). This is the AC Josephson effect. It may be mentioned that the passage of the AC current is accompanied by emission of radiation of the same frequency $2eV_1/\hbar$; this frequency is ~ 100 GHz.

Fig. 13.37 Current-voltage characteristic for a DC Josephson junction according to (13.87)



13.7.3 Macroscopic Quantum Interference Effect

The quantum interference between two Josephson tunnel junctions results in a current flow even in the absence of voltage difference. The superconducting quantum interference device (SQUID) is used to study this effect. The principle of SQUID is discussed in this section.

We consider a closed circuit consisting of a ring of superconducting material (Fig. 13.38) with arms a and b . The current entering through i_1 passes through the arms a and b with Josephson junctions X. It is then collected at i_2 . Let the phase difference for the two paths be ϕ_a and ϕ_b . In the absence of a magnetic flux within the ring, the two phases are equal. When a magnetic flux Φ is allowed to pass through the space enclosed by the ring, there will be a phase difference between the two paths. This will allow a current flow J_a and J_b in the two arms.

The phase difference in the presence of Φ is given by $\phi_b - \phi_a = (2e/\hbar c)\Phi$ or,

$$\phi_b = \phi_0 + \frac{e}{\hbar c}\Phi \text{ and } \phi_a = \phi_0 - \frac{e}{\hbar c}\Phi. \quad (13.88)$$

where ϕ_0 is the initial phase. The total current due to the phase difference is given by

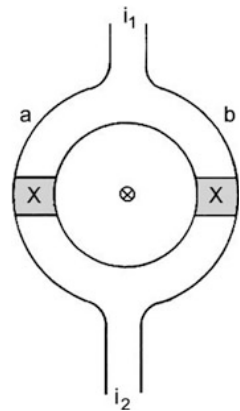
$$J_{\text{total}} = J_a + J_b = J_0 \left[\sin\left(\phi_0 + \frac{e}{\hbar c}\Phi\right) + \sin\left(\phi_0 - \frac{e}{\hbar c}\Phi\right) \right]. \quad (13.89)$$

Hence

$$J_{\text{total}} = 2(J_0 \sin \phi_0) \cos \frac{e\Phi}{\hbar c}. \quad (13.90)$$

Equation (13.90) shows that J_{total} varies with Φ and has maxima when $(e\Phi/\hbar c) = n\pi$ where n is an integer. The current flowing in the two junctions

Fig. 13.38 Principle of a DC SQUID



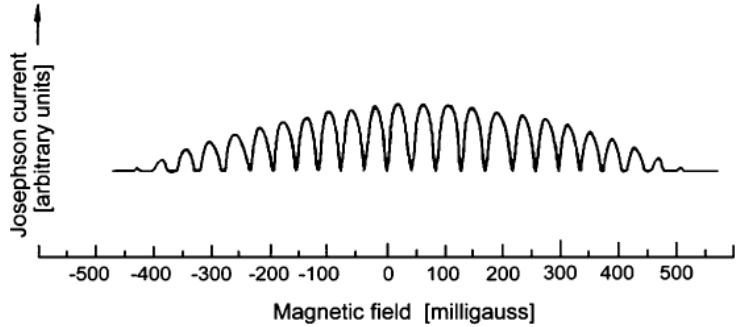
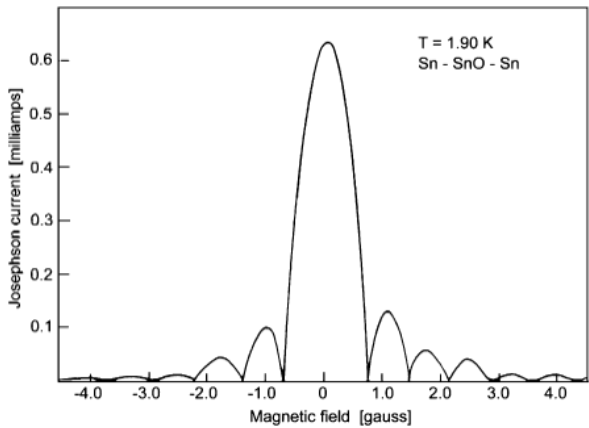


Fig. 13.39 Plot of Josephson current versus magnetic field for a two-junction interferometer showing interference effect

Fig. 13.40 Plot of Josephson current versus magnetic field for a single junction showing diffraction effect



oscillates with a period equal to a flux quantum. The oscillations are the result of interference between macroscopic wave functions at the two junctions. This is just like the interference between two coherent light beams. A long period variation is also observed which is due to the finite dimensions of the two junctions giving rise to diffraction effects.

The Josephson effects are manifestations of quantum interference phenomena on a macroscopic scale. The term quantum refers to the quantum state of the superconductors and the term interference to the properties due to difference in phase of a particular state. The experimental traces of Josephson current versus magnetic field recorded using Sn–SnO–Sn tunnel junctions are shown in Fig. 13.39 [41] and Fig. 13.40 [42]. Figure 13.39 shows the interference effects recorded with two junctions; the envelop arises because of finite size of individual junctions. Figure 13.40 shows diffraction effect recorded with a single junction.

Fig. 13.41 Comparison of currents due to Josephson tunneling, quasiparticle tunneling and normal tunneling types; the Shapiro steps are due to absorption of microwave field

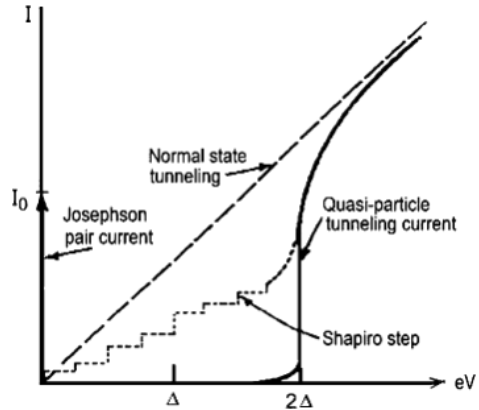


Figure 13.41 shows various tunneling currents and Shapiro steps, which are induced when a Josephson junction is exposed to microwave radiation, i.e. the radiation is absorbed at step interval of $\Delta V = h\nu / 2e$ [43].

13.7.4 Electrical Characteristics of a SQUID

We have discussed the principle of DC SQUID in the earlier section. We now see the basic electrical characteristics. When the flux through the SQUID is changed the current-voltage characteristics change. The squid is a very sensitive detector of the change in flux. A SQUID is essentially a flux-to-voltage transducer that converts a change in magnetic flux to a change in voltage easily detectable with conventional electronics. The schematic and electrical characteristics of a DC SQUID is shown in Fig. 13.42. In Fig. 13.42a the DC SQUID is biased with a current I and flux Φ . Each of the two Josephson tunnel junctions X and X is shunted with the self capacitance C and resistance R . The inductance of the loop is L . The flux and current bias are selected such that the voltage is almost linear in applied flux. Figure 13.42b shows the current-voltage characteristics with flux $\Phi = n\Phi_0$ and $(n + \frac{1}{2})\Phi_0$. The plot in Fig. 13.42c shows the voltage across the SQUID as a function of the flux (Φ/Φ_0) at a fixed bias (the dashed line in Fig. 13.42b).

Thus when the flux through the SQUID is changed it is recorded as a change in the voltage. Hence the SQUID is called a sensitive magnetometer. It can record extremely feeble magnetic fields of the order of 10^{-15} T.

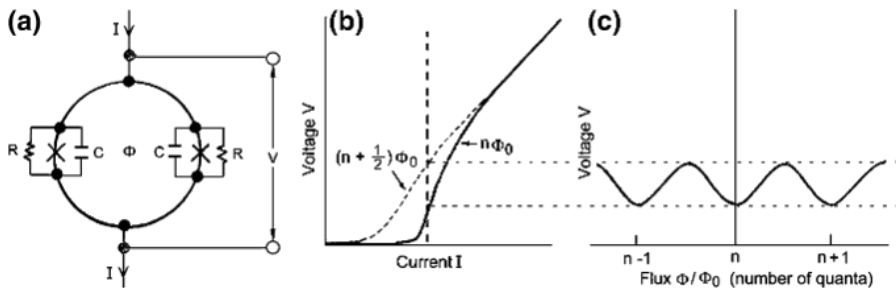


Fig. 13.42 Schematic and characteristics of a DC SQUID

13.8 High Temperature Superconductors

13.8.1 Structure and Transition Temperature

Till 1973, the highest temperature for superconducting phase transition T_c was ≈ 23 K for Nb_3Ge . The discovery of superconductivity in the family of cuprates by Bednorz and Muller [12] heralded the beginning of the era of high temperature superconductivity (HTS). These are La_2CuO_4 in which lanthanum is partially replaced by barium or strontium. Since then a large number of Cu oxides with $T_c \approx 30$ K were discovered. In early 1987, several investigators found that replacing lanthanum with yttrium raised the critical temperature to 92 K. This was the first superconductor of the type $\text{YBa}_2\text{Cu}_3\text{O}_7$ to show superconductivity with $T_c > 77$ K (liquid nitrogen temperature). It was a major step in the development of HTS technology as liquid nitrogen can be produced cheaply from air. The compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ (YBCO) is referred to as 123 superconductor. $\text{YBa}_2\text{Cu}_3\text{O}_7$ has an orthorhombic structure. The structure has two Cu–O sheets in the ab plane and Cu–O chains along the b axis. The structure of $\text{YBa}_2\text{Cu}_3\text{O}_7$ and the variation of resistance with temperature is shown in Fig. 13.43. Most of the high T_c superconductors have tetragonal or nearly tetragonal (some have small distortion leading to orthorhombic) structure. The materials can be generally considered as quasi-tetragonal.

Around 1988, a transition temperature of $T_c = 125$ K was reached in the compound $\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{13}$. This was the beginning of high temperature superconductor technology era with possible commercial applications. Figure 13.44 shows year-wise record of T_c of some materials. The highest recorded T_c is 173 K for $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_8$ at 30 GPa pressure.

Several compounds with the general formula $\text{La}_{2-x}\text{M}_x\text{CuO}_4$ are HT superconductors. They have the tetragonal K_2NiF_4 structure at room temperature and becomes orthorhombic around 180 K (Fig. 13.45a). The lanthanum ion can be replaced by Pr, Nd or Gd ions; M represents either Sr or Ba. The temperature variation of resistivity of some of these compounds is shown in Fig. 13.45b.

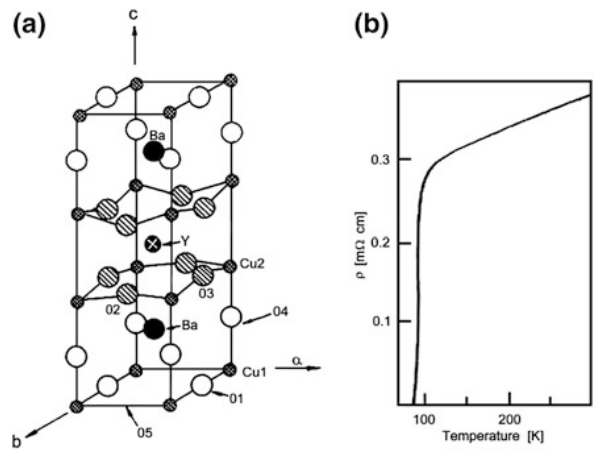


Fig. 13.43 YBa₂Cu₃O₇: **a** structure, **b** temperature variation of resistance

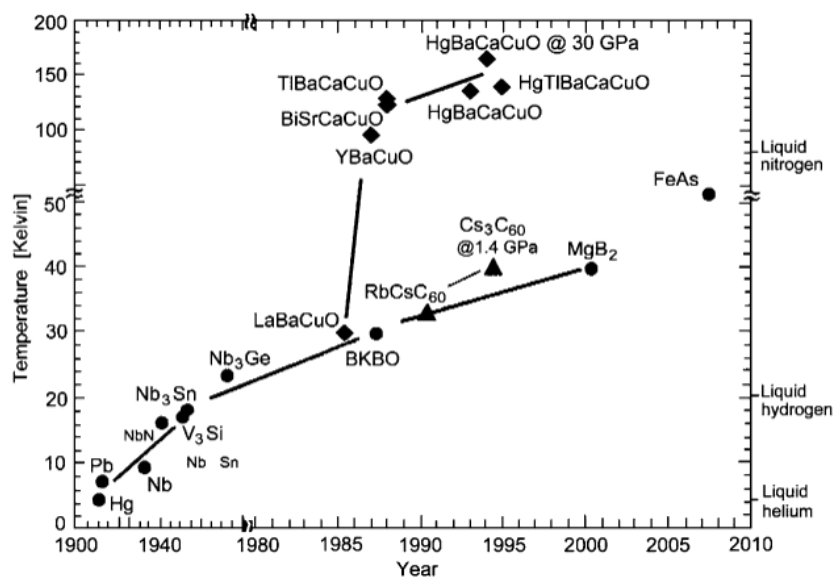


Fig. 13.44 Year-wise discovery of superconductivity

A list of some superconductors along with their T_c values and the number (n) of adjacent Cu–O planes in the unit cell are shown in Table 13.8. It is to be noted that larger the number of Cu–O planes the higher are the T_c values within a group. This is true up to a value of $n \geq 3$. It is also to be seen that for $n = 1, 2$ and 3 , values of T_c approximately lie in the range 20–85 K, 60–92 K and 105–125 K.

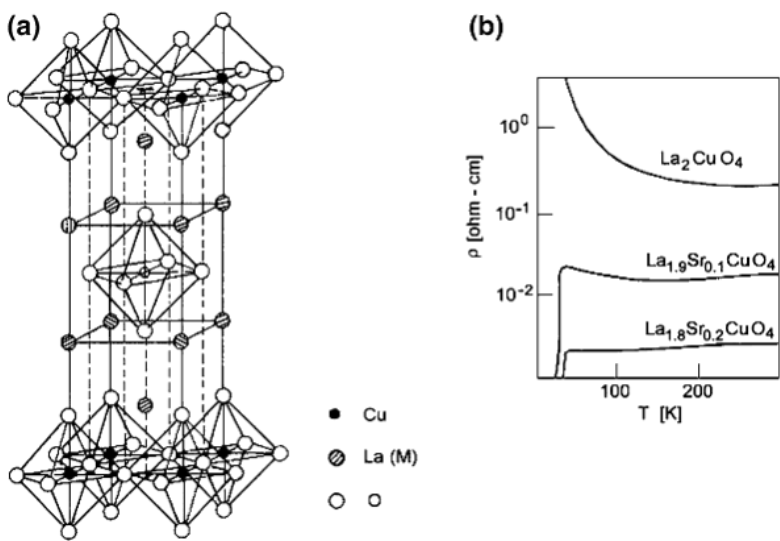


Fig. 13.45 $\text{La}_{2-x}\text{M}_x\text{CuO}_4$: **a** structure, **b** temperature variation of resistance

Table 13.8 Transition temperatures (T_c) and number n of Cu–O planes in some HT superconductors

Superconductor	T_c (K)	n
$\text{Tl}_2\text{Ba}_2\text{CuO}_6$	80	1
$\text{Tl}_2\text{Ba}_2\text{CaCuO}_8$	108	2
$\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	125	3
$\text{Bi}_2\text{Sr}_2\text{CuO}_6$	20	1
$\text{Bi}_2\text{Sr}_2\text{CaCuO}_8$	85	2
$\text{Bi}_2\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}_{10}$	110	3
$\text{TlBa}_2\text{CuO}_5$	50	1
$\text{TlBa}_2\text{CaCu}_2\text{O}_7$	80	2
$\text{TlBa}_2\text{Ca}_2\text{Cu}_3\text{O}_9$	110	3
$\text{TlBa}_2\text{Ca}_3\text{Cu}_4\text{O}_{11}$	122	4

The Cu–O planes are responsible for many of the electrical properties of the HTS materials and also the structures. The anisotropic structures are the cause of anisotropic properties.

13.8.2 Properties

- (i) The HTS materials are all type II superconductors with two critical magnetic fields H_{c1} and H_{c2} .
- (ii) The transition temperature can be varied by changing the doping.
- (iii) The presence of two-dimensional Cu–O sheets in the ab plane and Cu–O chains along the perpendicular direction shows that the structures of HTS materials are highly anisotropic. This is reflected in many of the properties.

The important length scales are the coherence length and magnetic penetration depth. These are related to H_{c1} and H_{c2} by the relations

$$H_{c1} = \Phi_0 / 4\pi\lambda^2$$

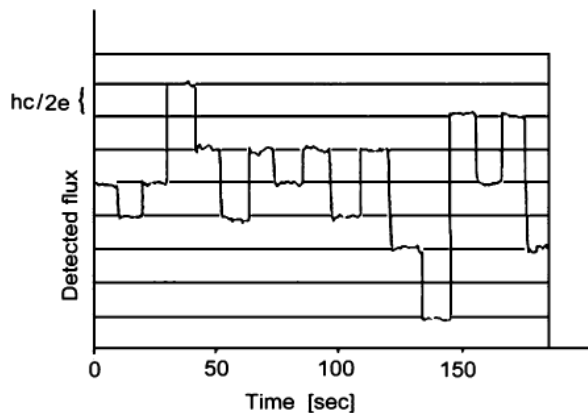
and

$$H_{c2} = \Phi_0 / 2\pi\xi^2. \quad (13.91)$$

For Y123 single crystal, values of the penetration depth λ in the ab plane and along the c axis at 0 K are $\sim 1,500$ and $\sim 7,000$ Å respectively. Values of the coherence length ξ in the ab plane and along the c axis at 0 K are 12–16 Å and 1.3–3 Å respectively. These are estimated from experimental results on magnetic fields using (13.91).

- (iv) Experiments have shown that the magnetic flux Φ is quantized in units of $\Phi_0 = hc/2e$ as in the case of conventional superconductors. This has been measured for Y123 superconductor. Magnetic flux through a ring of Y123 superconductor was obtained by using a ring of outer and inner diameters 10 and 5 mm respectively. The magnetic flux was measured as a function of time. The flux generated by electromagnetic noise below T_c was measured using a SQUID magnetometer. Figure 13.46 shows flux versus time. The flux

Fig. 13.46 Flux versus time in a Y123 superconducting ring



was calibrated with a long solenoid passing through the ring. The value of fluxoid quantum was found to be $\Phi_0 = (0.97 \pm 0.04) hc/2e$ [7].

- (v) Results on energy gap obtained from tunneling, infrared reflectivity and specific heat measurements indicate anisotropy in the energy gap. The results show large difference between the gaps in the c and ab directions. For Y123, IR reflectivity measurements have given values for the energy gap as $(2\Delta/k_B T_c) \approx 8$ in ab plane and ≈ 3.5 along the c direction.
- (vi) Experimental evidence for electron pairing has been established. The mechanism of electron pairing is not apparent.

13.9 Applications

There are numerous applications of superconductivity. Only some of them are mentioned here.

Power Transmission: For large scale applications, superconductors are advantageous over normal metals, because of low resistance i.e. low power loss. But the cost involved in production of low temperatures and in handling the superconducting materials make the conventional materials more advantageous. On the other hand the use of HTS materials eliminates iron cores in motors, generators and transformers resulting in reduction in size, weight and losses from iron cores. Also, HTS materials are of type II superconductors with high magnetic fields. These properties make them particularly useful.

Levitation: Superconductors display the property of levitation. This has led to the magnetically levitated high speed trains using liquid helium. These trains 'MAG-LEV' have been tried in Japan, Germany and USA.

SQUIDS: Early applications of superconducting materials in electronics were and continue to be instruments based primarily on superb sensitivity in magnetic fields. This has led to several small-scale applications where the currents involved are small. These are Josephson junction devices, infrared detectors and SQUID magnetometers.

Biomagnetism: An important area where superconductor devices are used is in the field of biomagnetism. Josephson junctions are fabricated with HTS materials for use in medical diagnostics such as checking the function of heart and brain. This technique is known as magnetic resonance imaging (MRI). By impinging a strong superconductor-derived magnetic field onto the human body, hydrogen atoms that exist in the body's water and fat molecules are forced to accept energy from the magnetic field. They then release this energy at a frequency that can be detected and displayed graphically by a computer. MRI was actually discovered in the mid-1940s but the first MRI on a human body was performed only in 1977. It took almost five hours to produce one image. The time taken nowadays is only a few seconds.

Satellite Communication: Besides being ultra-sensitive detectors of magnetic fields, superconductors also excel in detection of extremely faint electromagnetic

signals e.g. signals originating from outer space. SQUID detectors of high sensitivity are used in many radioastronomy observatories worldwide. HTS thin films have been fabricated into passive microwave devices such as filters and resonators for use in space technology.

The most significant change in satellite communication is the ‘all-digital receiver’ on a one centimeter superconducting niobium chip. The chip contains about 11,000 Josephson junctions laid out to form superconducting Rapid Signal Flux Quantum (RSFQ) circuits.

Computer Technology: In the field of computer technology, low temperature technology can take computing speed far beyond the theoretical limits of silicon while simultaneously effecting major reductions in size and power requirements. Another emerging area is in the field of communication. One of them is the superconducting router of high speed data communication up to 160 GHz. Since internet traffic is increasing exponentially, superconductor technology is expected to meet this requirement.

The development of RSFQ logic circuits has ultra-fast operating speed of several hundred GHz and very low power loss (nanowatts).

Other Developing Areas: Superconducting X-ray detectors and ultra fast superconducting light detectors.

13.10 Concluding Remarks

Superconductivity is one of the most important areas in physics. A variety of experimental observations have been made like the Meissner effect, the isotope effect and the Josephson effects. In order to explain the experimental observations, a number of theories were formulated like Londons’ theory, the BCS theory and the Ginzburg–Landau theory. These theories involved the introduction of new concepts like the energy gap, the coherence length, the critical field and the Cooper pairs. Over the years a number of applications of superconductivity have been discovered like superconducting magnets, maglev trains and SQUID devices.

The importance of superconductivity is indicated by the award of the Nobel Prize to a large number of investigators in this field. These are:

The 1913 Nobel Prize to Kammerlingh Onnes for his investigations of properties of matter at low temperatures.

The 1972 Nobel Prize to Bardeen, Cooper and Schrieffer for their theory of superconductivity.

The 1973 Nobel Prize to Giaever for the discovery of tunneling in superconductors and to Josephson for the discovery of phenomena known after him.

The 1987 Nobel Prize to Bednorz and Muller for the discovery of high temperature superconductors.

The 2003 Nobel Prize to Abrikosov and Ginzburg for contributions to the theory of superconductivity.

13.11 Problems

1. The critical temperature of mercury with isotopic mass 199.5 atomic mass units is 4.185 K. Calculate its T_c when the isotopic mass is changed to 203.4.
2. A superconducting metal has a critical temperature of 7.26 K at zero magnetic field and a critical field of 8×10^5 A/m at 0 K. Find the critical field at 4 K.
3. Calculate the ratio of penetration depths at 0 and 3.6 K for lead with critical temperature of 7.22 K.
4. Calculate the frequency of radiation emitted from a Josephson junction; the voltage across the junction is 5 μ V.
5. From thermodynamic considerations show that the entropy of a superconductor in the normal state is higher than the entropy in the super conducting state.
6. Describe the salient features of BCS theory and the BCS ground state.
7. Describe the working of a DC SQUID and mention some applications.

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